SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	1 - Introduction & Basic Concepts

1. Introduction to Surveying

Surveying is the art and science of determining the relative positions of points on, above or beneath the surface of the earth by measuring horizontal distances, vertical distances (elevations) and angles, and then plotting them to scale as a plan or map.

- **Plan:** a horizontal projection of the area drawn to a large scale (small areas, e.g. 1 cm = 5 m).
- **Map:** a drawing of a large area drawn to a small scale (e.g. 1 cm = 1 km).
- Surveying is the first step of every civil engineering project - roads, railways, canals, dams, buildings, bridges, pipelines.
- Measurements are made in either the **horizontal plane** (distances, angles, bearings) or the **vertical plane** (levels, heights).

Primary divisions

- **Plane surveying:** earth surface taken as a plane; curvature ignored. Used for areas up to about 260 sq.km.
- **Geodetic surveying:** curvature of the earth is considered. Used for large areas and high accuracy (more than 260 sq.km).

2. Objectives of Surveying

- To prepare a **topographic map** showing natural and man-made features - hills, rivers, roads, towns.
- To prepare a **cadastral map** showing field and property boundaries, used for land records and taxation.
- To prepare **engineering maps** for roads, railways, canals, dams etc.; to fix the alignment of the project.

- To calculate **areas and volumes** of land and earthwork (cutting and filling).
- To **set out** (mark on ground) the proposed structure from its plan – this is called setting out or layout.
- To collect data for estimating cost, and to resurvey and check existing works.

3. Classification of Surveying

(a) Based on the nature of the field

- **Land surveying:** topographical, cadastral, city and engineering surveys.
- **Marine / hydrographic surveying:** water bodies – tides, depth (soundings), shore lines.
- **Astronomical surveying:** positions of heavenly bodies to fix latitude, longitude and azimuth.
- **Aerial surveying:** photographs taken from aircraft or drones (see Unit V).

(b) Based on the instrument used

Survey	Main instrument	Used for
Chain survey	Chain / tape, cross staff	Small, open, flat areas
Compass survey	Prismatic compass	Bearings of lines, traverse
Plane table survey	Plane table, alidade	Plotting in the field itself
Levelling	Dumpy / tilting level, staff	Elevations, profiles
Theodolite survey	Theodolite	Precise angles, traverse
Tacheometry	Tacheometer	Rapid horizontal / vertical dist.
Total station / GPS	EDM, GNSS receiver	Modern high-precision work

(c) Based on the method

- **Triangulation:** control points form a network of triangles; one baseline and angles are measured.
- **Traversing:** a series of connected lines whose lengths and directions are measured.

(d) Based on the purpose

- Engineering survey (reconnaissance, preliminary, location), military survey, mine survey, archaeological survey, geological survey.

4. Principles of Surveying

KEY PRINCIPLES:

Two fundamental principles: (1) Work from the whole to the part. (2) Locate a new station by at least two independent measurements from fixed reference points.

Principle 1: Working from whole to part

First establish a major **control** (a framework of well-spaced, accurately measured main stations). Then fill in the minor details inside this frame using less accurate methods.

- Reason: errors do not accumulate; any error in small details stays confined to that portion.
- The reverse (part to whole) lets errors build up and the final result gets distorted.

Principle 2: Fixing a new point

A new station can be fixed from two reference points A and B by any one of these:

- Measuring two distances (linear measurements) from A and B.
- Measuring one distance and one angle (or bearing).
- Measuring two angles (or bearings) - intersection.
- Measuring distance to the new point from A and a perpendicular offset from the line AB.

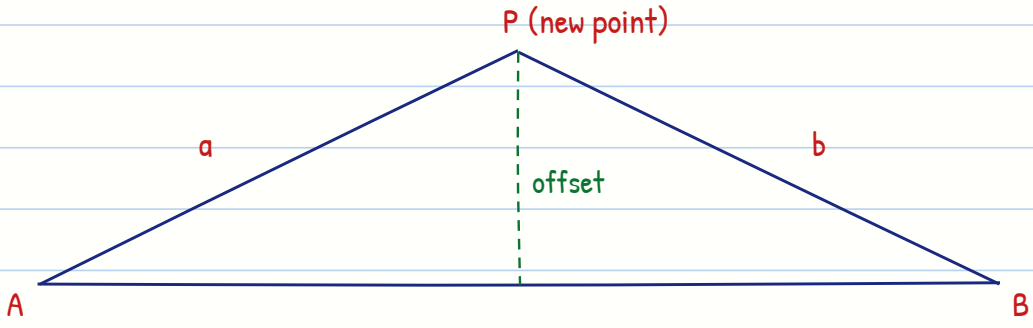


Fig 1: fixing a point P from baseline AB

Other general principles: (3) Check lines and tie lines must always be provided to detect errors. (4) Accuracy of the work should suit the purpose and scale (precision should not exceed what the drawing can show).

5. Surveying Accessories

Accessory	Purpose
Chain / Tape	Direct measurement of linear distance
Arrows (marking pins)	10 per set; mark every chain length on ground

Ranging rods	2-3 m poles, painted red-white 0.2 m bands; to mark stations and align the line
Ranging poles	Like rods but 3-6 m long, for long lines
Offset rod	3 m rod with hook, used to measure offsets
Cross staff	Sets out right angles (offsets) - open, French, adjustable
Optical square	Sets right angles using two mirrors at 45 deg
Plumb bob	Transfers points vertically; plumbing a staff
Pegs	Wooden, mark station points; 2.5 cm sq, 15 cm long
Levelling staff	Reading of levels with a level

6. Introduction to Compass Surveying

A compass survey fixes the ***direction (bearing)*** of survey lines using a magnetic needle. Lengths are measured with chain or tape. It is suited to rough, rapid work on large areas, rugged or wooded country.

- Instruments: ***prismatic compass*** (more common) and surveyor's compass.
- Method: open ***traverse*** (not closed) or ***closed traverse*** . Fore bearing and back bearing of each line are taken.
- Accuracy: angular accuracy about 15 minutes; not suitable near steel structures, power lines.

7. Introduction to Levelling

Levelling is the branch of surveying that determines the relative heights (elevations) of points above or below a reference surface called the ***datum*** (usually Mean Sea Level).

- Instruments: dumpy level, tilting level, auto level, levelling staff.
- Uses: preparing longitudinal and cross sections, contouring, fixing grades and formation levels for roads, canals, drains.
- Reduced Level (R.L.) = elevation of a point above datum. Detailed study in Unit II.

8. Introduction to Plane Table Surveying

In ***plane table surveying*** fieldwork and plotting are done simultaneously. A drawing board (plane table) is mounted on a tripod and a sheet of paper fixed on it. Points are sighted through an ***alidade*** and rays are drawn directly.

- Accessories: plane table with tripod, alidade (plain or telescopic), spirit level, plumbing fork with plumb bob, trough compass, drawing sheet, pins.
- Methods: ***radiation, intersection, traversing, resection*** .
- Advantages: no field book needed, mistakes and omissions noticed on the spot, suited for small scale maps.
- Disadvantages: not suitable in wet or windy weather; needs more carrying equipment; less accurate.

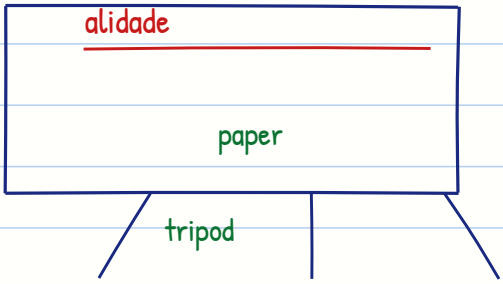


Fig 2: plane table on tripod

9. Linear Distances – Approximate Methods

Used for reconnaissance and checking of precise methods:

- ***Pacing:*** number of paces x average pace length (about 0.75 m). Accuracy about 1 in 100. Pace length is found by walking a known distance.
- ***Passometer:*** a watch-like device that counts steps; distance = steps x step length.
- ***Pedometer:*** records distance directly for a set stride length.
- ***Odometer / speedometer:*** attached to a wheel or vehicle; distance = wheel turns x circumference. Used on roads.
- ***Measuring wheel (perambulator):*** wheel pushed along the ground; dial shows distance.
- ***Chain-less estimation:*** by time taken (speed x time), or by sound/optical range finders.
- ***Tacheometric / EDM*** methods are accurate and are covered separately.

10. Direct Methods of Measuring Distance

Distance is directly measured on the ground by chains or tapes.

Types of chains

Chain	Length	No. of links	Remarks
Metric chain	20 m or 30 m	100 or 150 (20 m = 100)	Link = 0.2 m; tallies at every 1 m
Gunter's chain	66 ft	100	Link = 7.92 in; 10 chains = 1 furlong
Engineer's chain	100 ft	100	Link = 1 ft
Revenue chain	33 ft	16	Used for cadastral survey (land records)

- Metric chain: tallies – ***brass tags*** – used for easy reading: 1 tooth = 2 m, 2 teeth = 4 m, 3 teeth = 6 m, 4 teeth = 8 m; centre tag (10 m) is round.
- Chain is made of galvanised mild steel wire, 4 mm dia; links bent at both ends and joined with three small rings; brass handles at ends.
- ***Length of chain*** is measured from outside of one handle to outside of the other.

Chain surveying: principle and basics

- Principle: ***triangulation*** – the area is divided into well-conditioned triangles; sides are measured, details by offsets.
- ***Base line*** = longest main line; ***check lines / proof lines*** verify accuracy; ***tie lines*** fix intermediate details.
- Well-conditioned triangle: no angle less than 30 deg or more than 120 deg (ideally equilateral).

Tapes

Tape	Material / features
Cloth or linen tape	Linen with brass wires; 10, 20, 30 m; stretches and shrinks – only for offsets
Metallic tape	Linen with fine brass/copper wires; more durable
Steel tape	Steel band 6–16 mm wide, 30 or 50 m; accurate; rust-prone
Invar tape	36% nickel + 64% steel alloy; very low thermal expansion; for base line (highest accuracy)

11. Ranging

Ranging is the process of establishing intermediate points on a straight line between two end stations so that the chain can be laid in a straight line.

Direct ranging

Intermediate points are visible from end stations. Ranging rods are fixed at A and B. The assistant at C (near A) is directed by the surveyor standing 2 m behind A and sighting A–B until all rods are in line.

Indirect (reciprocal) ranging

Used when end stations are not intervisible due to a rise or obstacle. Two chainmen M and N stand with rods at intermediate points, each in sight of the other's rod and one end. They direct each other alternately until both rods are in line with A and B.

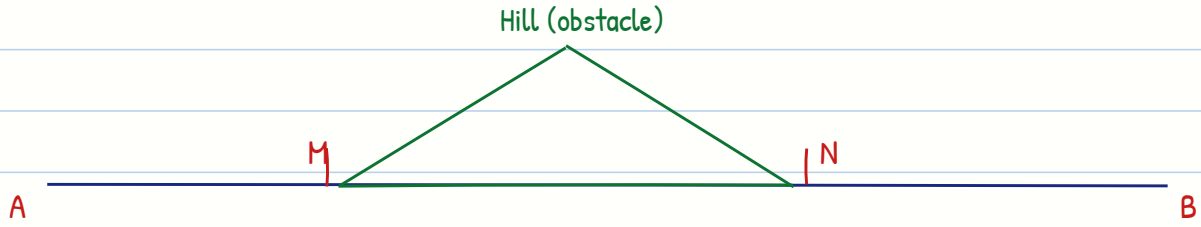


Fig 3: reciprocal ranging over a rise

12. Chaining a Line and Offsets

- The **leader** carries arrows and rods; the **follower** holds the rear handle and directs the leader on line.
- At the end of each chain the leader plants an arrow; the follower collects it. Arrow count in follower's hand = chains measured.
- **Offsets** are lateral measurements from the chain line to a feature. Perpendicular offset: set with cross staff or optical square. Oblique offset: measured at an angle (tie line).
- Limit: perpendicular offset up to 15 m (ordinary) and oblique offsets 20 m with chain; use of tape allows longer.

13. Errors in Chaining and Tape Corrections

Corrections are applied when the tape is not of standard length or conditions differ from standard. Errors may be cumulative (+/-) or compensating.

(a) Correction for wrong length (incorrect chain)

True length = L' x (L + e) / L

Correction Ce = L x e / L' (for measured length l: Ce = l x e / L)

e = change in chain length

- If the chain is **too long** , measured distance is too short -> **add** correction.
- If the chain is **too short** , measured distance is too long -> **subtract** correction.
- Area correction: True area = measured area x (L'/L)^2.

(b) Correction for temperature

Ct = alpha x (Tm - To) x L

alpha = coefficient of expansion (steel: 11.2 x 10^-6 per deg C)

- If Tm > To (standard temp.), tape lengthens: correction is **added** ; otherwise subtracted.

(c) Correction for pull (tension)

Cp = (P - Po) x L / (A x E)

P = applied pull, Po = standard pull, A = area of cross-section, E = Young's modulus

(d) Correction for sag (tape suspended between supports)

Cs = w^2 x l^3 / (24 x P^2) per span, or (W^2 x L)/(24 x n^2 x P^2)

- w = weight of tape per metre, W = total weight, n = number of equal spans. Sag correction is **always negative** .
- Sag is reduced by increasing the pull or the number of supports. Normal tension: correct pull P = 0.204 x W x sqrt(AE) / sqrt(Pm - Po) (for no-sag case).

(e) Correction for slope

Cslope = h^2 / 2L (approx.) or L (1 - cos theta)

Horizontal distance D = sqrt(L^2 - h^2) = L cos theta

- h = difference in level between the two ends, L = slope distance. Always **negative** .

(f) Correction for alignment, (g) correction to MSL

Calign = d^2 / 2L (d = deviation from line)

Cmsl = - L x h / R (h = mean elevation, R = 6370 km)

MEMORISE:

Sag, slope and alignment corrections are always SUBTRACTIVE. Temperature and pull may be + or -, wrong length depends on whether the tape is long or short.

Worked example 1

A 30 m chain was found to be 0.05 m too long. A line measured 450 m. Find true length.

True length = 450 x (30 + 0.05)/30 = 450.75 m

Worked example 2 (slope)

A line of 120 m on a uniform slope has a height difference of 4 m. Find horizontal distance.

C = h^2/2L = 16/240 = 0.067 m

D = 120 - 0.067 = 119.933 m

Errors in chaining (summary)

- **Personal errors** - wrong reading, careless marking, bad ranging.
- **Natural errors** - temperature, wind, sag, slope.
- **Instrumental errors** - wrong length, bent links, loose rings.
- Mistakes: miscounting chain lengths, misreading tallies, wrong booking of figures.
- Remedies: standardise the chain before use, check it with a test gauge, apply corrections.

14. Prismatic Compass

A ***prismatic compass*** is a portable magnetic instrument used to measure the bearing of a line. It has a magnetic needle on an agate cap, graduated aluminium ring (card) and a prism to read the bearing.

Parts

- ***Box*** (non-magnetic), with glass cover and lid; ***magnetic needle*** with agate cap on a pivot.
- ***Graduated ring*** (0 to 360 deg, graduated inverted; 0 at the south end).
- ***Object vane*** with hair, ***eye vane*** with slit and ***prism*** , sunshades (coloured glasses), lifting pin and brake pin.

Temporary adjustments

- ***Centring:*** place the compass over the station using plumb bob.
- ***Levelling:*** keep the box horizontal so that the card swings freely (ball and socket).
- ***Focusing the prism:*** move prism up or down until the graduations are sharp.

Observation

Sight the object vane to the ranging rod. Look through the slit, read the graduation seen against the hair line (through prism) when the needle stops.

15. Bearings and Included Angles

- ***Meridian:*** fixed reference direction – true, magnetic or arbitrary.
- ***True meridian:*** line through geographical North Pole and South Pole. True bearing = azimuth.
- ***Magnetic meridian:*** direction of freely suspended magnetic needle. Magnetic bearing.
- ***Arbitrary meridian:*** any convenient direction (e.g. the first line of the survey).
- ***Grid meridian:*** direction parallel to the central meridian of a grid system.

Bearing

Bearing of a line is the horizontal angle between the meridian and the line, measured clockwise from the North.

- ***Fore bearing (F.B.):*** bearing taken in the direction of progress of the survey.
- ***Back bearing (B.B.):*** bearing in the opposite direction.

B.B. = F.B. +/- 180 deg
(add 180 if F.B. < 180; subtract 180 if F.B. > 180)

Included angle

Interior angle between two successive lines at a station.

Included angle = F.B. of next line - B.B. of previous line
(add 360 deg if the result is negative)

CHECK:
For a closed traverse of n sides: sum of interior angles = (2n - 4) x 90 deg. Check this to find observation errors.

16. Systems of Bearing: W.C.B. and Q.B.

Whole Circle Bearing (W.C.B.)

Bearing measured **clockwise from North** from 0 to 360 deg. A single value fixes the line. Used in the prismatic compass.

Quadrantal Bearing (Q.B.)

Bearing measured from North or South, **whichever is nearer** , towards East or West, with value from 0 to 90 deg. Used in the surveyor's compass. Written as N 40 E or S 25 W.

W.C.B. range	Quadrant	Q.B. formula
0 - 90	I (NE)	N (W.C.B.) E
90 - 180	II (SE)	S (180 - W.C.B.) E
180 - 270	III (SW)	S (W.C.B. - 180) W
270 - 360	IV (NW)	N (360 - W.C.B.) W

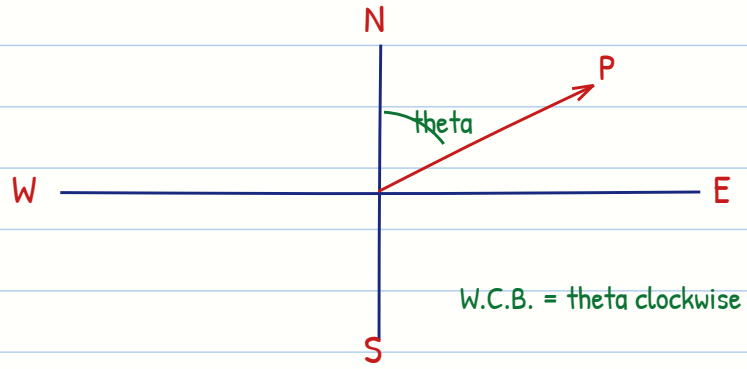


Fig 4: whole circle bearing

Examples

- F.B. of AB = 125 deg 30' -> B.B. = 305 deg 30'. Q.B. of AB = S 54 deg 30' E.
- W.C.B. 245 deg -> Q.B. = S 65 W. W.C.B. 320 deg -> N 40 W.
- Q.B. N 30 W -> W.C.B. = 330 deg; S 20 E -> W.C.B. = 160 deg.

17. Local Attraction

The magnetic needle gets deflected from the magnetic meridian due to nearby magnetic substances - iron objects, steel structures, rails, cables, magnetite rocks. This is ***local attraction*** .

- Detection: if F.B. and B.B. of a line differ by exactly 180 deg, both stations are free from attraction. Otherwise, there is local attraction at one or both ends.
- Correction methods: (1) start from a line whose F.B. and B.B. differ by 180 deg and take that as correct; (2) distribute the error among the lines using the included angle method.
- ***Method 1:*** the observed bearings of lines at an affected station are all wrong by the same error; correct by applying the same error.
- ***Method 2 (angle method):*** the included angles at all stations are not affected by local attraction. Compute them and correct the bearings of all lines from a correct bearing.

Sample

FB of AB = 60 deg 30', BB = 240 deg 30' (difference 180). So A and B are free. FB of BC = 120 deg 10', BB of CB = 301 deg 40'. The difference is 181 deg 30', so C has attraction: correction at C is -1 deg 30' applied to all bearings measured at C.

18. Magnetic Declination and Dip

Magnetic declination (D)

Horizontal angle between the true meridian and the magnetic meridian. It varies with place and time (secular, annual, diurnal and irregular variations).

- ***East declination*** – magnetic north lies to the east of true north (+ve). ***West declination*** – magnetic north to the west (-ve).

> True bearing = Magnetic bearing + declination
> (declination East = +, West = -)

- Lines of equal declination = ***isogonic lines*** ; lines of zero declination = ***agonic lines*** .
- Variations: secular (long-term, ~ 100–200 yr), annual (about 1–2 min), diurnal (changes through the day, max about 1 deg), irregular (magnetic storms).

Dip of the needle

The angle that the magnetic needle makes with the horizontal at a place. The needle's north end dips down in the northern hemisphere and the south end in the southern hemisphere.

- Dip is 90 deg at the magnetic poles and 0 deg at the magnetic equator. To balance it, a ***rider (small brass coil)*** is placed on the needle's higher end.
- Position of the rider shows which hemisphere the survey is in. This is done by the manufacturer for each country.

19. Quick Revision Points

- Whole to part, two independent measurements.
- Corrections: sag, slope, alignment always subtract.
- B.B. = F.B. +/- 180. Included angle = FB(next) - BB(previous).
- W.C.B. 0–360 from North; Q.B. 0–90 from N/S towards E/W.
- Local attraction: FB – BB difference not equal to 180 deg. Declination: true = magnetic + declination (E +).

SURVEYING

Handwritten Study Notes

Subject : SURVEYING
Code : 23A01301T
Unit : II - Levelling, Contouring, Areas, Volumes

I. Levelling - Introduction

Levelling determines the relative elevations of points. It is done by measuring vertical distances with respect to a horizontal line of sight.

Terms used in levelling

Term	Meaning
Level surface	Surface everywhere normal to the plumb line (e.g. still water surface); curved
Horizontal plane	Plane tangential to the level surface at a point
Datum	Arbitrary reference level; Mean Sea Level (MSL) is the standard datum
Reduced Level (R.L.)	Elevation of a point above or below the datum
Benchmark (B.M.)	Fixed point of known R.L. (GTS, permanent, arbitrary, temporary)
Line of collimation	Line passing through the cross hairs and optical centre of the objective
Axis of telescope	Line joining optical centre and the centre of the eyepiece
Height of Instrument (H.I.)	Elevation of line of collimation above datum
Back sight (B.S.)	First staff reading after setting the level; on a point of known R.L.
Fore sight (F.S.)	Last reading before shifting the instrument
Intermediate sight (I.S.)	Any reading between B.S. and F.S. on the same setting

Change point (C.P.)	Point where both F.S. and B.S. are taken on shifting the instrument
Station	Point where the staff is held

2. Types of Levels

- **Dumpy level:** telescope rigidly fixed to the supports; permanent adjustment; simple and stable; most commonly used.
- **Wye (Y) level:** telescope rests in two Y-shaped supports and can be rotated about its axis, removed and reversed end to end.
- **Tilting level:** telescope can be tilted a little about a horizontal axis by a tilting screw; bubble is brought central for each reading.
- **Automatic (self-aligning) level:** a compensator keeps the line of sight horizontal after rough levelling by a circular bubble; fast and accurate.
- **Digital level:** reads bar-coded staff electronically; data stored automatically.
- **Cooke's reversible level, Cushing's level, Hand level, Abney level, Laser level** – other types for special uses.

Dumpy level – main parts

Telescope (objective, eyepiece, diaphragm with cross hairs), level tube (bubble tube), tribrach with three foot screws, tripod, and a clamp and tangent screw.

Levelling staff

- Wooden or aluminium rod, 3–5 m long, graduated in metres, 0.1 m and 0.005 m (5 mm) divisions. Types: solid, folding, telescopic.
- Self-reading staff (e.g. Sop-with, Philadelphia, Enhanced) and target staff are used.

3. Temporary Adjustments of Level

- **Setting up:** fix the tripod at convenient height with the head approximately level; fix the level on it.
- **Levelling up:** with foot screws, bring the bubble to the centre using the rule of two screws at a time and then the third, in two perpendicular positions.
- **Elimination of parallax:** (a) focus eyepiece until the cross hairs are distinct and black; (b) focus the objective to bring the staff image on the plane of cross hairs.

PARALLAX:
Parallax is present when the image of the staff and cross hairs do not lie in the same plane: the reading changes when the eye moves up or down.

4. Methods of Levelling and Determination of Levels

(A) Height of Instrument (H.P.) Method

H.I. = R.L. of B.M. + B.S.
R.L. of point = H.I. – (I.S. or F.S.)

- Suited to a large number of intermediate points (contouring, cross sections); fewer calculations.
- Arithmetic check: ***Sum of B.S. – Sum of F.S. = Last R.L. – First R.L.***

(B) Rise and Fall Method

Rise or fall = (previous reading) – (next reading)
If positive -> rise; negative -> fall
R.L. = previous R.L. + rise (or – fall)

- Used in differential levelling and longitudinal sections; checks all points, including I.S.
- Check: ***Sum B.S. – Sum F.S. = Sum Rises – Sum Falls = Last R.L. – First R.L.***

COMPARE:
Rise & Fall method has complete check (including intermediate sights). H.I. method has no check on the I.S. readings and is quicker with many I.S.

Sample level book

Stn	B.S.	I.S.	F.S.	H.I.	R.L.
BM	1.525			101.525	100.000
A		1.810			99.715
B	0.890		2.040	100.375	99.485
C			1.250		99.125

Check: sum B.S. = 2.415; sum F.S. = 3.290; difference = -0.875; last R.L. – first R.L. = 99.125 – 100.000 = -0.875. OK.

Types of levelling

- **Simple levelling:** one setting, B.S. and F.S. on two points.
- **Differential levelling:** difference of level between two far-apart points; several set-ups and change points.
- **Fly levelling:** differential levelling to connect a B.M. to the work, only B.S. and F.S.
- **Check levelling:** done to check the earlier work, ends on a starting B.M.
- **Profile (longitudinal) levelling and cross sectioning:** levels along the centre line and at right angles to it for roads, canals.
- **Reciprocal levelling:** used for crossing a river or a valley; eliminates collimation, curvature and refraction errors.
- **Precise levelling:** high-precision, with invar staff, for geodetic control.

5. Reciprocal Levelling

Two instrument positions, one on each bank (near A, then near B). Readings are taken on both staves from both positions.

True difference in level: $h = [(a1 - b1) + (a2 - b2)] / 2$

Combined error $e = [(a1 - b1) - (a2 - b2)] / 2$

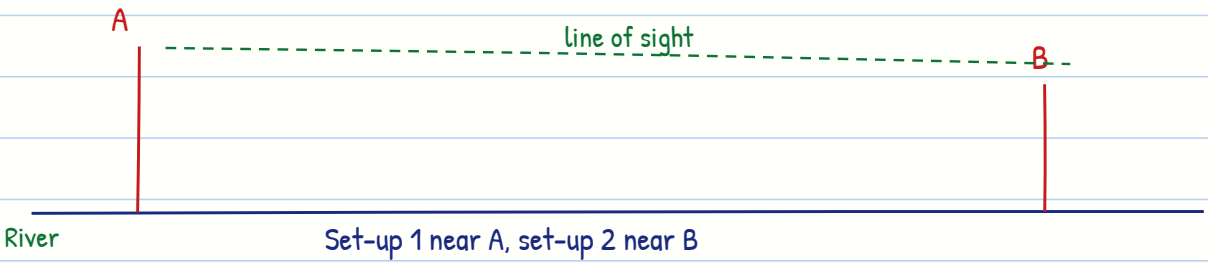


Fig 1: reciprocal levelling

6. Effect of Curvature and Refraction

A level line is curved (follows the earth); the line of sight is straight. So readings at a distance are too high. Also the atmosphere bends light rays downward, which reduces the error a little.

Curvature correction

$$Cc = d^2 / 2R \quad \text{(always subtractive)}$$
$$\text{With d in km: } Cc = 0.0785\, d^2 \text{ metres}$$

Refraction correction

$$Cr = d^2 / 14R = 0.0112\, d^2 \text{ metres} \quad \text{(additive)}$$
$$(Cr = 1/7 \text{ of } Cc)$$

Combined correction

$$C = Cc - Cr = 0.0673\, d^2 \text{ metres} \quad \text{(subtractive)}$$
$$d \text{ in km; } R = 6370 \text{ km}$$

REMEMBER:
Sight at distance d (km): staff reading appears too high by $0.0673\, d^2$ m. So subtract this from the staff reading. For d = 1 km -> 6.73 cm.

Distance of visible horizon (Dip)

$$\text{Dip: } h = 0.0673\, D^2 \quad (h \text{ in m, } D \text{ in km})$$
$$D = \text{sqrt}(h / 0.0673) = 3.855\, \text{sqrt}(h) \text{ km}$$

Example: an observer's eye is at 20 m above sea. Visible horizon: D = 3.855 x sqrt(20) = 17.2 km.

Sources of error in levelling

- **Instrumental:** line of collimation not parallel to the axis of bubble tube; staff graduation error; imperfect parallax adjustment.
- **Personal:** mistakes in reading, booking, holding the staff not vertical, careless levelling.
- **Natural:** curvature, refraction, wind, temperature, settlement of tripod.
- Staff not vertical gives readings too large; ***rocking the staff*** and taking the minimum reading removes this.
- Remedies: keep B.S. and F.S. distances equal, check adjustments (two-peg test), avoid long sights.

Two peg test (collimation error)

Two pegs A and B about 60–100 m apart. Set up level midway (reading a1, b1: correct difference). Then set up near A, take readings a2 and b2. If (a2 – b2) = (a1 – b1) the line of collimation is horizontal; otherwise adjust the diaphragm until the reading b2′ = a2 – (a1 – b1).

7. Contouring

A ***contour*** is an imaginary line joining points of equal elevation on the ground. A ***contour map*** shows the shape of the ground by such lines.

- ***Contour interval:*** vertical distance between two consecutive contours. Depends on scale, relief, purpose and the time and cost permitted.
- ***Horizontal equivalent:*** horizontal distance between two consecutive contours; varies with the slope.
- Typical intervals: 0.5 to 1 m for small scale area, 2–5 m for rugged ground, 10–20 m for a big map.

Characteristics of contours

- Two contours of different elevations ***never intersect*** (except at overhanging cliffs or vertical cliff where they merge or cross).
- Contours of the same elevation never cross; every contour closes on itself within or outside the map.
- Closely spaced contours indicate ***steep slope*** ; widely spaced contours indicate a ***gentle slope*** ; equal spacing means ***uniform slope*** ; parallel, straight, equally spaced lines indicate a plane surface.
- A contour crosses ridge or valley lines at ***right angles*** .
- ***Ring contours*** with higher values inside denote a hill; with lower values inside denote a depression.
- ***U-shaped (convex)*** contours pointing to lower ground represent ***ridges*** ; ***V-shaped*** pointing to higher ground represent a ***valley*** .
- A contour cannot end abruptly; it cannot split into two.

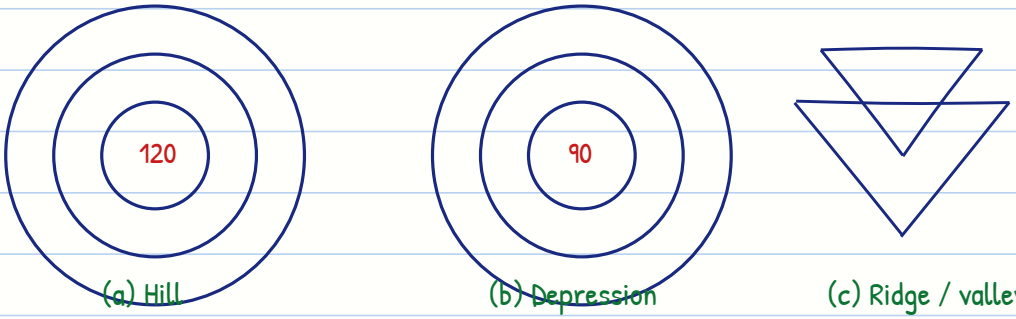


Fig 2: hill, depression and ridge contours

Uses of contour maps

- To study the nature of the ground and choose suitable sites for roads, canals, dams, railways.
- To determine the ***intervisibility*** of two points and draw a section.
- To compute the ***volume of earthwork*** and reservoir capacities, catchment area.
- To trace out ***gradient lines*** (e.g. a road at a given gradient).
- To find the ***drainage and watershed*** areas.

8. Methods of Contour Surveying

Direct method

The contour points are directly located on the ground by levelling, and then surveyed and plotted. Accurate but slow and costly; suited to small areas where accuracy is needed.

- A level is set; B.S. on the B.M. gives the H.I. Required staff reading for the contour of elevation R is: staff reading = H.I. – R.
- The staff man moves up/down the slope until the required reading is obtained; point marked by a peg and then fixed by chain/plane table.

Indirect method

Spot levels of selected points are taken, and contour positions are interpolated. Quicker and cheaper. Mainly for hilly and large areas.

Indirect method	How it works
Squares (grid) method	Area divided into squares (5–20 m); levels at corners; contours interpolated

Cross-section method	Series of lines at right angles to a central line; levels along each
Radial line method	Radial lines from a central point; levels at known distances; best for hills / small areas
Tacheometric method	Stadia hairs give elevations and distances; for hilly regions

Interpolation of contours

- **Estimation:** by eye; rough and only for small works.
- **Arithmetic calculation:** proportions of the level difference and the horizontal distance.
- **Graphical method:** tracing cloth with equally spaced parallel lines; fix the points at the given levels; the lines cut at the intersections give contour points.

Example of arithmetic interpolation: points A (RL 101.4) and B (RL 103.2) are 36 m apart. Contour 102 m is at (102 − 101.4)/(103.2 − 101.4) × 36 = 12 m from A.

9. Areas – Regular and Irregular Boundaries

Areas from field notes (cross-staff survey)

- **Triangle:** A = sqrt[s(s-a)(s-b)(s-c)] where s = (a+b+c)/2; or A = 1/2 b x h.
- **Rectangle:** L x B. **Trapezoid:** 1/2 (a + b) x h. **Circle:** pi r^2.

Areas with an irregular boundary – offsets

Trapezoidal rule:

$$A = d\left[\frac{O_0 + O_n}{2} + O_1 + O_2 + \ldots + O_{n-1}\right]$$

d = equal interval between offsets; O0, O1, … On = the offsets (ordinates). Assumes the boundary between offsets is a straight line.

Simpson's one-third rule (n even, i.e. odd number of ordinates):

$$A = \frac{d}{3}\left[(O_0 + O_n) + 4(O_1 + O_3 + \ldots) + 2(O_2 + O_4 + \ldots)\right]$$

CONDITION:
Simpson's rule: number of ordinates must be ODD (even number of strips). Boundary assumed to be a parabola; more accurate than trapezoidal. If not odd, apply Simpson to the first strips and the trapezoidal rule to the last one.

- **Average ordinate rule:** A = (sum of ordinates / number of ordinates) x length of base.
- **Mid-ordinate rule:** A = d x (sum of mid-ordinates).

Area from coordinates (closed traverse)

2A = Sum [Xi (Y(i+1) - Y(i-1))] = Sum of products of X and Y cross

- **Planimeter:** mechanical instrument: tracing the boundary gives area; Area = M (final - initial reading + 10 N when zero is crossed). **Digital planimeter** gives area directly.
- Area by dividing into **triangles** from a map, or by **graph (squares)** method counting squares.

Example (trapezoidal and Simpson)

Offsets at 10 m interval: 3.0, 4.2, 5.1, 4.8, 3.6 (m). Trapezoidal: A = 10[(3.0+3.6)/2 + 4.2 + 5.1 + 4.8] = 10[3.3 + 14.1] = 174 sq.m.

Simpson: A = 10/3 [(3.0+3.6) + 4(4.2+4.8) + 2(5.1)] = 10/3 [6.6 + 36 + 10.2] = 176 sq.m.

10. Volumes - Earthwork

Volume of earthwork in **cutting** (excavation) and **embankment** (filling) is needed for estimating the cost of roads, railways and canals.

Level section (formation level horizontal, ground level across the section)

Area = (b + s h) h (b = formation width, s : 1 side slope, h = depth/height at centre)

Cross-section for a level section: A = b h + s h^2

Two-level and three-level sections

When the ground slopes across the section, the cross-section is split into triangles and trapezoids (using the side slopes to find where the slopes meet the ground), and the

areas are added. The side-slope toe distance on each side = s x (depth at that side).

Calculation of volume

Method	Formula
Trapezoidal (average end area)	V = d [(A1 + An)/2 + A2 + ... + A(n-1)]
Prismoidal formula	V = d/3 [(A1 + An) + 4 (A2 + A4...) + 2 (A3 + A5...)]
Prismoidal correction (level section)	Cp = (s d / 6)(h1 - h2)^2
Mean area method	V = (A1 + A2)/2 x d (small segments)

- ***Trapezoidal rule*** is simpler but gives ***larger*** (overestimate) volume; the prismoidal formula is more exact and needs an odd number of sections.
- Prismoidal correction is ***subtracted*** from the trapezoidal volume: V(prism) = V(trap) - Cp.

Example

Level section, formation width 10 m, side slope 2:1. Length 20 m; depth at A1 = 1.5 m, A2 = 2.5 m. Area A1 = (10 + 2 x 1.5) x 1.5 = 19.5; A2 = (10 + 2 x 2.5) x 2.5 = 37.5. Volume = (19.5 + 37.5)/2 x 20 = 570 cu.m.

II. Capacity of Reservoirs

The storage capacity of a reservoir at a site is obtained from a ***contour map*** of the area. Area enclosed by each contour (found with a planimeter) is computed, and the volume between contours is calculated.

Trapezoidal: V = h [(A1 + An)/2 + A2 + A3 + ... + A(n-1)]

Prismoidal: V = h/3 [A1 + An + 4 (A2 + A4 + ...) + 2 (A3 + A5 + ...)]

- h = contour interval; A1, A2... areas enclosed by successive contours from the bottom to FRL (full reservoir level).
- ***Cone formula*** between two contour areas: V = h/3 (A1 + A2 + sqrt(A1 A2)) - more accurate for gradually changing sections.
- Plot an ***elevation - capacity curve*** and an ***elevation - area curve*** to know the storage at any water level.

- Dead storage, useful storage and flood storage are the three zones considered.

12. Benchmarks and Field Procedure

- **G.T.S. benchmark:** Great Trigonometrical Survey benchmark, fixed with great accuracy, a permanent stone pillar with R.L. above MSL.
- **Permanent B.M.:** fixed by govt. departments (PWD, railways) on stable structures such as culverts, bridge piers, building plinths.
- **Arbitrary B.M.:** assumed R.L. (e.g. 100.000 m) used for small works where MSL is not known.
- **Temporary B.M.:** established at the end of a day's work, to continue the next day.

Field procedure of differential levelling

- Set up the level midway between B.M. and the first change point; level up; remove parallax.
- Take B.S. on the B.M.; then F.S. on the change point (the staff holder keeps the staff on the same point while the level is shifted).
- Move the level ahead, take B.S. on the same C.P., and continue to the last point.
- Keep B.S. and F.S. distances approximately equal; maximum sight length about 100 m.
- Close the work on a known B.M. (check levelling) and verify the closing error: allowable error = 12 sqrt(K) mm for ordinary work, K in km.

13. Worked Problems

Problem 1: Rise and Fall method

Staff readings on consecutive points: 2.285 (B.S. on B.M. of RL 200.000), 1.650, 2.100, 1.325 (F.S.), 1.860 (B.S.), 0.950 (F.S.). Find the RL of all points.

Pt	B.S.	I.S.	F.S.	Rise	Fall	R.L.
BM	2.285					200.000
1		1.650		0.635		200.635
2		2.100			0.450	200.185
3 (C.P.)	1.860		1.325	0.775		200.960
4			0.950	0.910		201.870

Check: sum B.S. = 4.145; sum F.S. = 2.275; difference = +1.870; sum rise − sum fall =

2.320 − 0.450 = +1.870; last RL − first RL = 201.870 − 200.000 = +1.870. All three agree.

Problem 2: Curvature and refraction

Find the combined correction and the true difference in level if the staff reading at 800 m distance is 2.450 m and the instrument height is 1.520 m.

$$C = 0.0673 \times (0.8)^2 = 0.0431 \text{ m}$$

$$\text{Corrected reading} = 2.450 - 0.043 = 2.407 \text{ m}$$

$$\text{True fall} = 2.407 - 1.520 = 0.887 \text{ m}$$

Problem 3: Gradient line from a contour map

To fix a road at gradient 1 in 25 with contour interval 2 m, the horizontal distance between two successive contours is 25 × 2 = 50 m. Set a divider to 50 m on the map scale and step from contour to contour.

$$\text{Horizontal equivalent} = \text{contour interval} \times (\text{gradient denominator})$$

$$= 2 \times 25 = 50 \text{ m}$$

Problem 4: Prismoidal formula

Three sections 30 m apart have areas 40, 52 and 60 sq.m. Find the volume.

$$\text{Prismoidal: } V = 30/3\,[(40 + 60) + 4 \times 52] = 10 \times 308 = 3080 \text{ cu.m}$$

$$\text{Trapezoidal: } V = 30\,[(40+60)/2 + 52] = 3060 \text{ cu.m}$$

Here the trapezoidal value differs by only 0.65%; in normal earthwork the difference is small, so trapezoidal volumes are usually accepted.

Problem 5: Reservoir capacity

Areas enclosed by contours at interval 5 m: 200, 1400, 3200, 5800, 9000 sq.m (with contour 0 at the bottom). Find the storage up to the top contour using the trapezoidal formula.

$$V = 5\,[\,(200 + 9000)/2 + 1400 + 3200 + 5800\,]$$

$$= 5\,[\,4600 + 10400\,] = 75{,}000 \text{ cu.m}$$

14. Planimeter – How the Area is Read

- A ***planimeter*** has a tracing arm, a pole arm, a roller (measuring wheel) with a graduated disc and a vernier.
- Place the pole outside the figure, trace the boundary clockwise with the tracer point, and note the initial and final reading of the wheel.

$$\text{Area} = M\,(F - I + 10N + C)$$

M = multiplying constant, N = no. of times zero mark passes the index,
C = constant of the instrument (when the pole is inside the area)

- Use: quick area of an irregular boundary or of contour-enclosed areas for reservoir capacity. Accuracy about 1 in 200.
- Verify by tracing a known area (e.g. circle or a square) to calibrate the planimeter.

15. Important Questions to Prepare

- Differentiate between rise and fall method and H.I. method with a neat level book.
- Derive the combined correction due to curvature and refraction.
- State the characteristics of contours; explain the direct and indirect methods of contouring.
- Explain the trapezoidal and Simpson's rules for area; derive the prismoidal formula.
- Compute the capacity of a reservoir from a contour map.

16. Quick Revision

- H.I. = R.L. + B.S.; R.L. = H.I. – F.S.; Check: sum B.S. – sum F.S. = last RL – first RL.
- Combined curvature and refraction: 0.0673 d^2 m (subtractive).
- Contours never cross (except cliff); steep = close; gentle = wide; closed rings = hill / depression.
- Trapezoidal rule, Simpson's 1/3 rule (odd ordinates), prismoidal formula (odd sections).

SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	III – Theodolite Surveying & Traversing

1. Theodolite – Introduction

A ***theodolite*** is the most accurate surveying instrument for measuring ***horizontal and vertical angles*** . It can also measure distances (tacheometry), prolong a line, and establish levels.

- Least count: 10, 20 or 30 seconds (vernier); 1 second for optical micrometre types.
- Uses: traversing, triangulation, setting out curves, tacheometric surveys, setting out buildings, heights of towers.

2. Types of Theodolites

Basis	Types
Transit / Non-transit	***Transit:*** telescope can be revolved through a complete vertical circle about its horizontal axis (commonly used). ***Non-transit:*** telescope cannot be transited (old).
Reading system	Vernier theodolite; micrometer (optical) theodolite; ***electronic / digital theodolite*** (reads angles by a display).
Size	Measured by the diameter of the lower plate: 8 cm, 10 cm, 12 cm, 15 cm and 25 cm.
Repeating / Directional	***Repeating:*** upper and lower plates can be clamped independently; for the repetition method. ***Directional (reiterating):*** only one plate; for the reiteration method.

Parts of a vernier transit theodolite

- ***Levelling head:*** tribrach with three foot screws; trivet; clamp to the tripod.
- ***Lower (scale) plate*** with the horizontal graduated circle; lower clamp and lower

tangent screw.

- ***Upper (vernier) plate*** with two verniers at 180 deg apart; upper clamp and upper tangent screw; plate levels.
- ***Standards (A-frame)*** support the ***telescope*** and the ***vertical circle*** (graduated), with verniers C and D on the T-frame (index arm) and a clip screw.
- ***Telescope*** with objective, eyepiece, diaphragm, focusing screw; ***altitude bubble*** ; plumb bob hook; compass (optional).

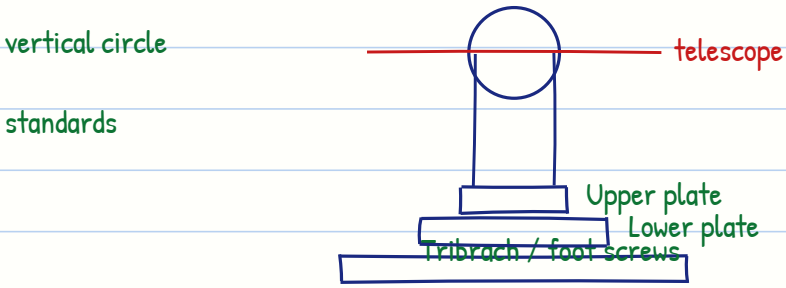


Fig 1: sketch of a transit theodolite

Technical terms

- ***Centring:*** keeping the vertical axis of the instrument exactly over the station mark.
- ***Transiting (plunging):*** revolving the telescope in a vertical plane through 180 deg about the horizontal axis.
- ***Face left / face right:*** vertical circle on left (face left) or on the right of the telescope (face right) when observing.
- ***Swinging the telescope:*** turning it in the horizontal plane. Right swing = clockwise; left swing = anticlockwise.
- ***Line of collimation:*** line through the cross hairs and the optical centre of the objective.
- ***Axis of the telescope, horizontal axis (trunnion axis), vertical axis*** - three axes that should be mutually at right angles.
- ***Telescope normal:*** vertical circle on the left, bubble of the telescope up. ***Inverted:*** vertical circle on the right, bubble down.

3. Temporary Adjustments of a Theodolite

Temporary adjustments are done at every station before taking observations:

- **Setting up:** place the tripod over the station with the head about level; fix the instrument; centre approximately by plumb bob and bring the plumb bob over the mark.
- **Centring:** shift the instrument on the tripod head (with the shifting head) until the plumb bob is exactly over the station.
- **Levelling up:** with the foot screws, bring the plate bubbles to the centre. Turn the upper plate until the plate level is parallel to two foot screws; centre the bubble; turn it 90 deg and centre using the third screw. Repeat till the bubble remains central in all positions.
- **Elimination of parallax:** (a) focus the eyepiece for distinct cross hairs, (b) focus the objective to see the object clear without any apparent movement of the cross hairs.

> **SEQUENCE:**
> Order: Setting up -> Centring -> Levelling -> Focusing the eyepiece and the objective. Permanent adjustments are done by the manufacturer / in the lab.

Permanent adjustments (relationship of axes)

- Plate bubble axis perpendicular to the vertical axis.
- Line of collimation perpendicular to the horizontal axis.
- Horizontal axis perpendicular to the vertical axis.
- Altitude bubble axis parallel to the line of collimation; vertical circle vernier reads zero when the line of sight is horizontal (index error nil).

4. Measurement of a Horizontal Angle

Ordinary method (single angle)

- Set up at station O and level. Set A vernier to zero (or any reading) using the upper plate; clamp the lower plate; sight A; fix with the lower clamp and tangent screw.
- Loosen the upper clamp; turn the telescope to sight B; clamp; bisect B with the tangent screw. Read both verniers; the mean difference from the initial reading gives angle AOB.

- Change face and repeat; the mean of the face-left and face-right values eliminates collimation and other errors.

Repetition method

Used to measure a horizontal angle **accurately** by **repeating** it several times and adding up on the circle. The final reading divided by the number of repetitions gives the angle. It reduces the least count error and the error of the instrument's graduation.

- Step 1: Measure angle AOB in the usual way with the upper clamp loose, lower clamp tight. Note the reading (e.g. a1).
- Step 2: Loosen the **lower** clamp, sight A again (reading stays the same), clamp lower; loosen upper, sight B. Reading is now a2 = twice the angle.
- Step 3: Repeat 3 times on face left, then change face and repeat 3 times on face right (total 6 repetitions).

Angle = (final reading – initial reading + 360 x number of full rotations) / number of repetitions

Least count error is reduced by n (i.e. 1/n of the original)

NOTE:

Repetition method needs a repeating (two-plate) theodolite. Errors due to poor graduation and eccentricity are eliminated; it cannot remove errors from slip, sighting, centring.

Example of observation booking

Station	To	Face	Vernier A	Vernier B	Mean	Angle
O	A	L	0 00 00	180 00 20	0 00 10	
	B	L	62 30 40	242 31 00	62 30 50	62 30 40
O	A	R	0 00 00	180 00 20	0 00 10	
	B	R	62 30 20	242 30 40	62 30 30	62 30 20

Mean angle = (62 30 40 + 62 30 20)/2 = 62 deg 30' 30".

Reiteration method

Used when **several angles at a station** are to be measured (angles AOB, BOC, COD, DOA). It is suited to a directional theodolite and avoids the accumulation of error in

measuring each separate angle.

- Set the instrument at O, level and centre. Sight the starting (reference) object A with a reading close to zero (first vernier 0 deg 00 00), the lower plate clamped. Read both verniers.
- Unclamp the upper plate, sight B, read; then C, D etc. in clockwise order, and finally close on A again (reading must agree with the initial reading within the least count – this is the closing check).
- Change face and repeat in the anticlockwise order (D, C, B, A). Take the mean of the two sets.
- To reduce errors, repeat in several ***sets*** with a different initial reading each time: 0, 60, 120 ... degrees; take the mean of corresponding angles.

5. Measurement of Vertical Angle

A ***vertical angle*** is the angle between the line of sight and the horizontal plane, measured in a vertical plane. It is an angle of elevation (+) or depression (–).

- Set up and level the theodolite at the station; bring the altitude bubble to the centre (the vertical circle verniers read zero when the line of sight is horizontal).
- Sight the object (use the horizontal cross hair); clamp the telescope; set the altitude bubble central by the clip screw; read both vertical verniers C and D.
- Change face and repeat; the mean of the readings eliminates the ***index error*** .

Vertical angle = (reading with the line of sight horizontal) – (reading at the object)

Index error = [(Reading face left) + (Reading face right) – 360 deg] / 2 ... (when read from 0 to 360)

Errors that cancel by changing face

- Error of collimation and horizontal axis; error of verticality of the vertical axis does not cancel.
- Eccentricity of verniers (by reading both verniers) and errors in graduation; index error of the vertical circle.

6. Trigonometrical Levelling

Trigonometrical levelling determines the difference in elevation of stations by the ***vertical angles*** and the horizontal distance (computed or measured). It is used in hilly regions, for heights of towers, chimneys, and for contour control.

(A) Base of the object is accessible

The instrument is set at station P; the horizontal distance D from P to the foot of the object Q can be measured directly. Let alpha = angle of elevation to the top, h = height of the instrument axis above the ground (= staff reading on B.M.).

Height of object H = D tan alpha (above the instrument axis)

R.L. of top = R.L. of P + h + D tan alpha

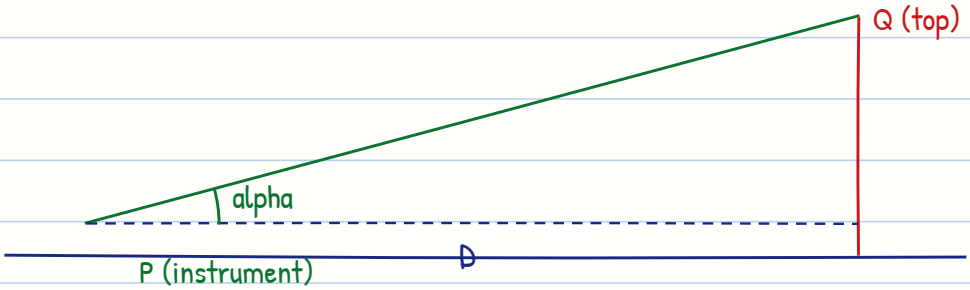


Fig 2: accessible base

(B) Base is inaccessible - instrument stations in the same vertical plane as th

The foot of the object is not accessible; two instrument stations P and Q are set in line with the object, a known distance b apart. Angles of elevation alpha1 at P (nearer) and alpha2 at Q (farther) are measured; the staff readings at B.M. are S1 and S2.

Case 1 (stations at the same level):

H = b tan(alpha1) tan(alpha2) / [tan(alpha1) - tan(alpha2)] = D1 tan alpha1

where D1 = b tan(alpha2) / [tan(alpha1) - tan(alpha2)]

Here P is nearer to the object with angle alpha1 > alpha2. D1 is the horizontal distance from P to the object. Then R.L. of object = R.L. of P (axis) + H.

Case 2 (axes of the two stations at different levels): find the height of the top

separately above each axis (using the staff readings S1 and S2 on the B.M. to get the R.L. of each axis); the two values of R.L. of the top should agree and the mean is taken.

(C) Base inaccessible – instrument stations NOT in the same vertical plane as

Two stations P and Q, distance b apart, are chosen such that P, Q and the object are not in a line. Horizontal angles at P and Q to the object are measured (angle alpha at P, angle beta at Q), and the vertical angles.

In triangle PQR: angle at R = 180 – (alpha + beta)

PR = b sin(beta) / sin(alpha + beta) (sine rule)

Height H = PR tan(vertical angle at P)

CORRECTION:

In trigonometrical levelling over long distances apply curvature and refraction correction: C = 0.0673 d^2 (m) to the computed height (add to the elevation of the far station).

Example

From a station P, the angle of elevation to the top of a chimney is 24 deg 15'. The base is accessible with a horizontal distance 85 m. The height of the instrument is 1.45 m.

H = 85 x tan 24 deg 15' = 85 x 0.4505 = 38.29 m

Total height above P ground = 38.29 + 1.45 = 39.74 m

7. Traversing

A ***traverse*** is a series of connected survey lines whose lengths and directions (angles or bearings) are measured. The survey is called ***traversing*** .

- ***Closed traverse:*** starts and ends at the same point (closed loop), or at two points of known position. Provides checks – used for boundary, plots, and control.
- ***Open traverse:*** does not close – used for long linear works like roads, canals, pipelines; has no checks.

Methods of traversing

Method	Description
Chain traversing	Angles by chain (triangles and tie lines); only for small areas

Chain and compass traversing	Bearings by compass; distances by chain. Also called free or loose needle method
Transit tape traversing	Angles by theodolite and distances by tape; for accurate work
Plane table traversing	Direction of lines by alidade and plot as the survey goes on
Theodolite traversing	Angles by theodolite; distances by tape or tacheometer or EDM

Theodolite traversing - methods of measuring angles

- ***By measuring included (interior) angles:*** angle between adjacent lines, usually in the anticlockwise direction for closed polygon (interior angles). Suits closed traverse, easily checked: sum = (2n - 4) x 90 deg.
- ***By direct observation of angles to the right (deflection angles):*** angle between the prolongation of the preceding line and the next line. Positive right, negative left. For open traverse - roads. Check: right deflection - left deflection = +/- 360 deg.
- ***By measuring bearings (magnetic) of lines:*** directly with the compass or with the theodolite fitted with a trough compass - loose needle method (angle from magnetic meridian at each station).
- ***Fast needle method (direct observation of bearing):*** the bearing of the first line is observed and then the angles; the bearings of other lines are computed.

8. Traverse Computations

Consecutive coordinates - Latitude and Departure

The ***latitude (L)*** of a line is its projection on the north-south axis (meridian); the ***departure (D)*** is its projection on the east-west axis.

Latitude L = l cos(theta) (N = +, S = -)

Departure D = l sin(theta) (E = +, W = -)

l = length of line; theta = W.C.B. of the line

W.C.B. (quadrant)	Latitude	Departure
0-90 (NE)	+	+

90-180 (SE)	-	+
180-270 (SW)	-	-
270-360 (NW)	+	-

Closing error

For a closed traverse, the algebraic sum of latitudes (Sum L) and of departures (Sum D) should each be zero. The non-zero sums give the **closing error** .

Closing error e = sqrt[(Sum L)^2 + (Sum D)^2]

Direction of error: tan(theta) = Sum D / Sum L

Relative precision = e / (perimeter) = 1 / (P / e)

Angular adjustment first

- Sum of interior angles of a closed polygon = (2n - 4) x 90 deg. The angular error = observed sum - theoretical sum.
- Distribute the angular error equally among all the angles (if equal weights) or in proportion to the weights.
- Recompute the bearings of the lines from the corrected angles; then compute L and D.

Balancing the traverse (adjusting the closing error)

Bowditch's rule (compass rule): used where angular and linear measurements are of the same accuracy. Correction in a line is proportional to its length.

Correction to latitude of a line = - (Sum L) x (l / Sum l)

Correction to departure of a line = - (Sum D) x (l / Sum l)

Transit rule: used when angular measurement is more precise than linear measurement (theodolite with taped distances).

Correction to latitude = - (Sum L) x (|L| / Sum |L|)

Correction to departure = - (Sum D) x (|D| / Sum |D|)

REMEMBER:
Sum l = perimeter. Sum |L| and Sum |D| = sum of the latitudes / departures without regard to the sign. Compass (Bowditch) rule is the most widely used.

Graphical method of adjusting (Bowditch)

Plot the traverse to scale as ABCDE (not closed, A and E' apart). Join A to E' and draw lines parallel to it, from B, C, D. The closing error is distributed along the traverse in proportion to the length of the sides (a straight line of cumulative lengths).

9. Gale's Traverse Table

A systematic tabular method of computing a closed traverse - includes the angles, bearings, lengths, latitudes, departures, corrections, adjusted values and the independent coordinates of each station.

Col.	Entry
1-2	Station; observed interior angle; correction; corrected angle
3	W.C.B. (from the preceding bearing + 180 - interior angle, etc.)
4	Length of line
5-6	Latitude and departure (+ and - columns)
7-8	Corrections by Bowditch; balanced latitudes and departures
9	Independent coordinates (N, E) of the stations (Northing = previous + L)

- Independent coordinates: Northing N = N(prev) + latitude; Easting E = E(prev) + departure.
- Choose the origin so that all the stations have positive coordinates (the most south-westerly station as origin).
- Area of the traverse can be determined from the coordinates (see Unit II).

10. Worked Example - Traverse Computation

A closed traverse ABCDA: AB = 100 m (N 0), BC = 80 m (E 90), CD = 100 m (S 180) and DA = 79 m (W 270). Check: Sum L = 100 - 100 = 0; Sum D = 80 - 79 = +1.

Closing error e = 1.0 m; perimeter = 359 m
Relative precision = 1/359

Bowditch: correction to departure of each line = -(+1.00) x l / 359. So AB = -0.28, BC

= -0.22, CD = -0.28, DA = -0.22; total = -1.00 m, which balances the +1.00 m error. Latitudes need no correction here (Sum L = 0).

II. Omitted Measurements (Introduction)

Sometimes the length or the bearing (or both) of one or two lines of a closed traverse cannot be measured in the field. These ***omitted measurements*** are computed from the condition that the traverse closes: Sum L = 0 and Sum D = 0.

Omitted	Method of finding
Length of one line	Sum of the known L and D gives the missing L and D; length = sqrt(L^2 + D^2); bearing = tan^-1(D / L)
Bearing of one line	Closing line of all other lines; cosine / sine rule (triangle formed by the missing line and the closing line)
Length and bearing of one line	Closing line of the rest is the missing line - closing error of the rest of the traverse is reversed
Length of one line and bearing of another	Solve a triangle by the sine rule with the closing line of the known sides
Lengths of two lines / Bearing of two lines	Solve using the closing line and the triangle (two unknowns, two equations)

- General idea: assume the known lines add up to a closing line (Sum L, Sum D); the missing portion makes up the closure.

Missing line: Latitude = - Sum L (known lines); Departure = - Sum D (known lines)

Length = sqrt(L^2 + D^2); Bearing = tan^-1 (D / L), with the proper quadrant

LIMIT:

All the missing measurements of one line (length and bearing) can be found by this method. If more than two quantities are omitted, the traverse cannot be solved by itself.

12. Sources of Errors in Theodolite Work

- **Instrumental:** imperfect adjustments, eccentricity of the verniers and the circle, graduation errors; eliminated by face-left/face-right and reading both verniers.
- **Personal:** inexact centring, bisection errors, poor levelling, parallax, wrong reading or booking.
- **Natural:** unequal refraction, wind, temperature, settling of tripod, sun on one side of instrument.
- Remedies: shade the instrument, tighten the tripod legs, measure on both faces, repeat the angles and check the sum.

13. Quick Revision

- Theodolite angles: repetition (single angle, accurate), reiteration (many angles at a station).
- Interior angles sum = (2n - 4) x 90 deg; Latitude = l cos(theta); Departure = l sin(theta).
- Bowditch for equal accuracy; transit rule when angles are more precise.
- Trigonometrical levelling: accessible base H = D tan(alpha); inaccessible base - two-station method.
- Omitted measurements: use closure Sum L = 0, Sum D = 0.

SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	IV - Curves, Tacheometry & Modern Methods

I. Curves - Types and Necessity

A ***curve*** is a smooth transition between two straight lines (tangents) of a road, railway or canal alignment that change direction. Curves provide a gradual change of direction for safety, comfort and economy.

- Necessity: to change the direction of the alignment smoothly; to avoid sudden jerks; to allow safe design speed; to adapt to the topography.

Classification

Type	Description
Horizontal curves	Change of direction in the horizontal plane: simple, compound, reverse, transition (spiral), lemniscate
Vertical curves	Change of gradient in the vertical plane: summit (crest) and sag (valley) curves

- ***Simple curve:*** a single circular arc connecting two tangents.
- ***Compound curve:*** two or more simple arcs of different radii, curving the same way and meeting at a common tangent point (P.C.C.).
- ***Reverse curve:*** two arcs of the same or different radii, curving in opposite directions (S-shape); used in hilly roads and in crossovers on railways.
- ***Transition curve:*** a curve of varying radius (cubic parabola / spiral / lemniscate) inserted between a straight and a circular curve - gives gradual superelevation and centrifugal force. Not in the syllabus in detail.

2. Elements of a Simple Circular Curve

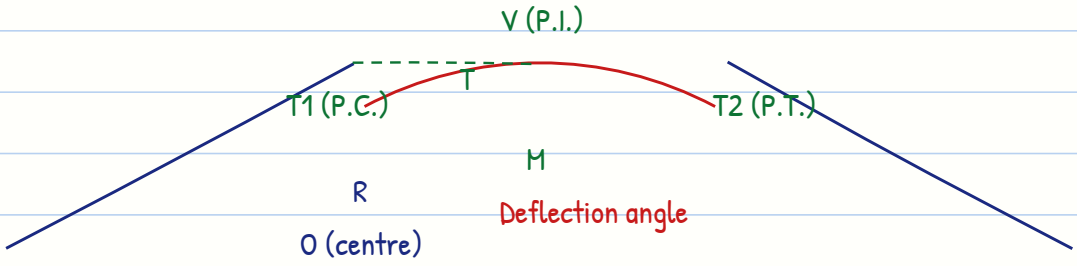


Fig 1: simple circular curve

Element	Meaning / formula
Tangents	Back tangent (AB) and forward tangent (BC) meeting at the point of intersection V (P.I.)
Deflection angle (D)	Angle by which the forward tangent deviates from the back tangent produced
Angle of intersection	180 deg - D
Point of curve (T1)	Beginning of the curve (P.C.)
Point of tangency (T2)	End of the curve (P.T.)
Tangent length (T)	T = R tan(D/2)
Length of curve (l)	l = pi R D / 180 = R x D (radians)
Long chord (L)	L = 2 R sin(D/2)
Apex distance	Ext. = R [sec(D/2) - 1]
Versed sine (M)	M = R [1 - cos(D/2)]
Degree of curve (Dc)	Angle subtended at the centre by a 30 m arc: R = 1718.9 / Dc

R = 1718.9 / Dc (metres, 30 m arc basis)

Chainage of T1 = Chainage of V - T

Chainage of T2 = Chainage of T1 + l

KEY:
Deflection angle at the apex = central angle of the curve. Tangents are equal: VT1 = VT2 = R tan(D/2). The length of the curve is the arc length T1 to T2.

Designation of curve

- **By radius:** the curve is specified by its radius, e.g. 300 m curve (used in roads).
- **By degree of curve:** angle subtended by a standard chord (30 m in metric railway practice, 20 m in India for railways).
- Sharpness of a curve: the smaller the radius (larger degree), the sharper the curve.

3. Setting Out a Simple Curve

Linear (chain and tape) methods:

- **Offsets from the long chord:** ordinates measured from the long chord at regular intervals; O_x = sqrt(R^2 - x^2) - (R - M), x measured from the mid-point of the chord; the central ordinate = M (versed sine).
- **Successive bisection of the arcs:** mid-point offsets are found by successive halving of the chord (versed sine).
- **Offsets from the tangents:** perpendicular offset O_x = R - sqrt(R^2 - x^2) (exact) or x^2 / 2R (approx.); radial offset O_x = sqrt(R^2 + x^2) - R.
- **Offsets from chords produced:** each peg is set with an offset from the chord produced; O1 = C1^2 / 2R, O2 = C2 (C1 + C2) / 2R.

Angular (instrumental) methods:

- **Rankine's method of deflection angles:** a theodolite at T1 sets pegs by the deflection (tangential) angle from the tangent; the most common method.
- **Two theodolite method:** two theodolites at T1 and T2 intersect at each peg; used where chaining is difficult (water, obstacles).
- **Tacheometric method and the total station method:** with polar coordinates.

Rankine's deflection angle method

The deflection (tangential) angle for a chord of length c is the angle between the tangent and the chord. It is half the central angle of the chord.

delta = (c / 2R) radians = 1718.9 c / R minutes

Total deflection angle at T2 = D / 2

First chord c1 (sub-chord), normal chords c = 30 m, last chord c2

- Deflection angle for the first sub-chord: delta1 = 1718.9 c1 / R (min); each normal chord: delta = 1718.9 c / R (min).
- Total deflection at the end of the curve = sum of the deflection angles = D/2. This is a ***check*** on the work.
- Procedure: set the theodolite at T1; with A at zero, sight V (tangent direction); turn by delta1; measure c1 from T1 to fix point 1; then turn by delta1 + delta2; swing the chain from peg 1 with length c; continue until T2.

Example

Two straights meet at chainage 1190 m with a deflection angle of 36 deg. Radius = 300 m. Find the tangent length, length of curve, and chainages of T1 and T2.

T = 300 tan 18 deg = 97.48 m

l = pi x 300 x 36 / 180 = 188.50 m

Chainage T1 = 1190 - 97.48 = 1092.52 m

Chainage T2 = 1092.52 + 188.50 = 1281.02 m

Obstacles in setting out curves

- When the point of intersection is inaccessible (e.g. in a lake or a building), the tangent lines are joined by a line connecting two points on tangents; the angles are measured and the deflection angle and the tangent length found by the sine rule.
- When the curve cannot be set out from one point because of an obstruction, the curve is set out from two or more stations (stage setting).

4. Compound Curve

A compound curve has ***two circular arcs*** with radii R1 and R2 in the same direction, with a common tangent at the point of compound curvature (P.C.C.). Elements: R1, R2, D1, D2, D (total deflection angle), the tangents VT1, VT2 and the common tangent t.

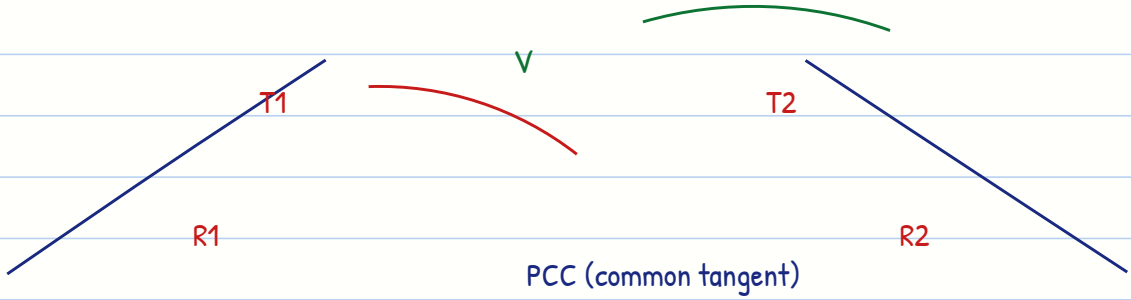


Fig 2: compound curve

$$D = D1 + D2$$

$$Ta = R1 \tan(D1/2);\ Tb = R2 \tan(D2/2);\ \text{common tangent } t = Ta + Tb$$

$$VT1 = Ta + t \sin(D2)/\sin(D);\quad VT2 = Tb + t \sin(D1)/\sin(D)$$

- Given D, R1, R2 and D1 (or D2) -> compute the rest. Used where the topography prevents a single curve; ***flat arc at the ends, sharper in the middle*** is preferred.
- Setting out: Rankine's method from T1 up to the point of compound curvature and then from T2 (back) or from the P.C.C. after orienting.

5. Reverse Curve

A reverse curve consists of two arcs of ***opposite curvature*** , joined at the ***point of reverse curvature (P.R.C.)*** where they have a common tangent. Used for connecting two parallel (or non-parallel) straights.

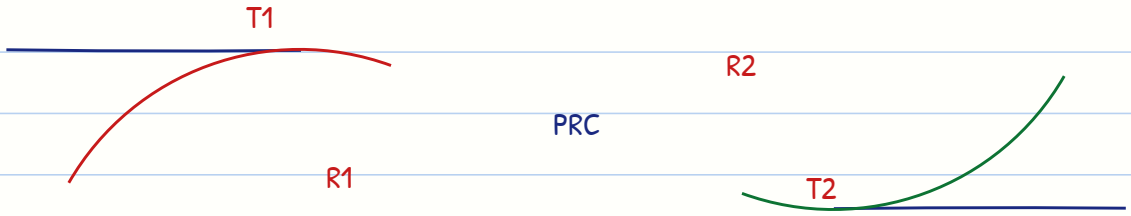


Fig 3: reverse curve between parallel straights

Two parallel straights a distance d apart, each arc subtending angle D:

$$d = (R1 + R2)(1 - \cos D)$$

If R1 = R2 = R: $d = 2R(1 - \cos D)$; length of curve = $2 \times \pi R D / 180$

- Common tangent at the PRC; the centres of the two arcs lie on the opposite sides of the common tangent.

- Reverse curves are ***not used*** in highways and fast railways because of the sudden change in the sign of superelevation (and the passengers feel discomfort); used for slow-moving roads, hills, yards.

> **COMPARE:**
> Compound curve: same side centres. Reverse curve: centres on opposite sides.
> Simple: one arc. Transition: varying radius.

6. Introduction to Tacheometric Surveying

Tacheometry is a rapid method of determining ***horizontal distances and elevations*** by optical means, using a ***tacheometer*** (a theodolite fitted with a stadia diaphragm) and a graduated staff. No chaining or levelling is needed.

- Used for: topographical surveys in hilly, difficult terrain; rough locations; checking chain surveys; contouring; traverse in rugged areas.
- Accuracy about 1 in 500 to 1 in 1000, lower than chaining.

Stadia method: principle

The ***stadia hairs*** (upper and lower) in the diaphragm intercept a length (staff intercept s) on a levelling staff at a distance D. The ratio D / s is constant for a given instrument.

Horizontal sight (staff vertical):

$$D = (f/i)\,s + (f + d) = K\,s + C$$

K = f/i = multiplying constant (usually 100), C = additive constant (usually 0 for an anallactic telescope)

- ***Anallactic lens:*** extra lens in the telescope that makes C = 0, so D = K s, simplifies the work.
- Here f = focal length of the objective, i = stadia hairs interval, d = distance from the objective to the vertical axis.

Inclined sight, staff vertical

Horizontal distance D = K s cos^2(theta) + C cos(theta)

Vertical distance V = (K s / 2) sin(2 theta) + C sin(theta)

R.L. of staff station = R.L. of instrument axis + V - h (for elevation)

(theta = angle of elevation; h = axial staff reading, i.e. middle hair)

- For angle of depression, V is negative: R.L. = R.L. (axis) - V - h.
- With C = 0: D = K s cos^2 theta; V = (K s / 2) sin 2 theta.

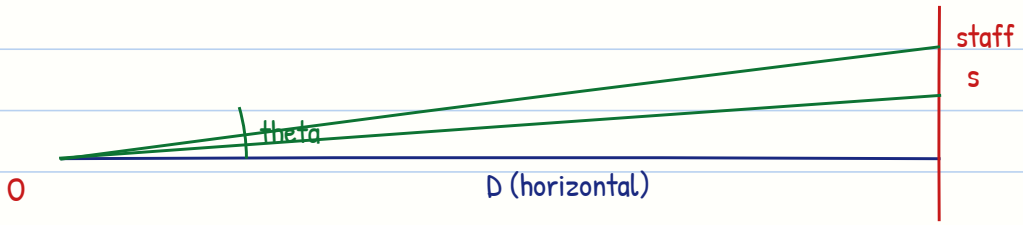


Fig 4: stadia tacheometry, inclined sight

Determination of the constants K and C

- Place the staff at two known distances D1 and D2 (e.g. 100 m and 200 m) from the instrument; read the intercepts s1 and s2.

K = (D2 - D1) / (s2 - s1)

C = D1 - K s1

Other systems of tacheometry

- ***Tangential method:*** staff with two targets at a known distance s apart; vertical angles to the targets give D = s / (tan alpha1 - tan alpha2).
- ***Subtense bar method:*** a horizontal bar of known length (2 m) at the station; the small horizontal angle subtended at the instrument gives distance D = (b/2) cot (beta/2).
- ***Moving hair method:*** stadia hairs can be moved; the movement is read from the screw head.

Example

A tacheometer with K = 100 and C = 0 is set at a station. At the staff reading with a vertical angle of +5 deg: upper hair 2.355, middle 1.905, lower 1.455. Staff intercept s

= 0.900 m.

D = 100 x 0.9 x cos^2 5 deg = 89.32 m
V = 100 x 0.9 / 2 x sin 10 deg = 7.81 m

7. Modern Surveying – EDM (Electromagnetic Distance Measurem

An **EDM** instrument measures the distance by the ***time of travel / phase difference*** of electromagnetic waves between the instrument and a reflector, in seconds. Quick, accurate (1 mm +/- 1 ppm), long range.

Principle

Distance D = (c x t) / 2 (pulse method)
c = velocity of light in the medium (approximately 3 x 10^8 m/s)

- Wave sent from the transmitter reflects at the target prism / reflector and returns; the ***time or phase difference*** is measured automatically and converted to distance.

Types of EDM instruments

Type	Wave / range
Microwave instruments (Tellurometer)	Microwaves; range up to 100 km; two instruments (master and remote); affected by weather less
Infrared (EDM) instruments	Infrared waves modulated; reflector prism; range 2-5 km; widely used with total station
Light wave (laser) instruments (Geodimeter)	Visible laser / light; range 2-3 km by prism; also reflectorless to 200-1000 m

- ***Phase comparison method:*** the phase difference between the sent and the received wave gives the fractional wavelength; several modulation frequencies resolve the integer ambiguity.
- ***Pulse method:*** the time of flight of short pulses is directly measured (used in reflectorless EDM).
- Corrections: atmospheric (temperature, pressure, humidity), instrument constant, prism constant, slope reduction, curvature, sea-level reduction.

8. Total Station

A ***total station*** combines an ***electronic theodolite*** , an ***EDM*** and a ***microprocessor*** with data storage in one instrument – it measures angles and distances and computes coordinates automatically.

- Parts: telescope with EDM, electronic angle encoder, microprocessor, keypad, display, data collector (memory card), battery, tribrach, prism reflector with pole.
- Measures horizontal angle, vertical angle, slope distance -> computes horizontal distance, vertical distance (height), and the ***northing, easting, elevation*** of the observed point.

Advantages

- Fast, accurate, and automatic data recording – minimum human error; transfer to a computer / CAD directly.
- Reflectorless measurement; long range; one instrument for many tasks; can do setting out; resection and free-station measurements.
- Reduced fieldwork and manpower; automatic correction for the curvature, refraction, temperature and pressure.

Applications

- Topographic and contour surveys; traversing and control; setting out of buildings, roads, bridges, tunnels; deformation monitoring of dams, buildings.
- Volume calculation (stockpile / earthwork), as-built surveys, mine surveys, cadastral and boundary surveys.

Field procedure (brief)

- Set up and level over the station; enter the instrument height, prism height, temperature, pressure; orient to a backsight; measure to the points; store and download.

9. Global Positioning System (GPS)

GPS is a ***satellite-based*** navigation and positioning system, operated by the USA, that gives the 3-D position (latitude, longitude, height) and time anywhere on or near the earth. ***GNSS*** is the general term for all such systems: GPS (USA), GLONASS (Russia), Galileo (Europe), BeiDou (China), ***NavIC (India)*** .

Three segments

- **Space segment:** 24+ satellites in 6 orbital planes at about 20,200 km altitude, period 12 hours; at least 4 satellites visible anywhere.
- **Control segment:** master control station and monitor stations that track the satellites, compute orbits and upload the data.
- **User segment:** receivers (handheld, geodetic) used by surveyors, navigators.

Working principle (trilateration)

Each satellite broadcasts its position and the time of transmission. The receiver measures the signal travel time and gets the distance (pseudo-range) = c x time. Distance from **at least 4 satellites** fixes x, y, z and the receiver clock error.

- **Static GPS:** two or more receivers occupy points for a long time; cm or mm accuracy. Used for control surveys.
- **Kinematic / RTK (Real Time Kinematic):** a base and a moving rover with radio link; 1-2 cm accuracy in real time.
- **DGPS:** corrections from a base station improve the accuracy to sub-meter.

Errors in GPS

- Satellite and receiver clock errors; orbit errors; ionospheric and tropospheric delay; multipath; poor satellite geometry (high DOP); selective availability (earlier).
- Applications: control surveys, geodetic surveys, cadastral and engineering surveys, navigation, deformation monitoring, GIS data collection, precision agriculture.

10. Drone (UAV) Survey

A **drone survey** uses an **Unmanned Aerial Vehicle (UAV)** carrying a camera (or LiDAR) to collect aerial images and data for mapping. Images are processed by **photogrammetry** software to give orthomosaics, point clouds, DEM, DSM and 3-D models.

- Workflow: flight planning -> ground control points (GCPs, measured by GPS) -> image capture with overlap (80 % forward and 60-70 % side) -> processing (aerial triangulation, dense cloud) -> outputs.
- Types: fixed wing (large areas, longer flights), multi-rotor (small areas, can hover) and hybrid VTOL.

- Advantages: quick, cheap, high resolution (1–5 cm per pixel), access to dangerous areas, frequent monitoring, safer for personnel.
- Limitations: battery life, weather, regulations (licenses and permits – India: Drone Rules 2021), accuracy depends on GCPs, vegetation reduces accuracy.
- Applications: topographic mapping, volume of stockpiles, road and railway corridor surveys, land records (SVAMITVA scheme), mining, construction progress monitoring, disaster assessment.

II. LiDAR Survey (Light Detection And Ranging)

LiDAR is an active remote sensing technique. A laser scanner emits rapid pulses of light (up to hundreds of thousands per second) towards the ground; the ***time of return*** of each pulse gives the distance; with the GPS position and the IMU (orientation) of the platform, each return is a 3-D point (x, y, z) – producing a ***point cloud*** .

Range R = (c x t) / 2

Point position = GNSS position + range vector rotated by IMU angles

- Platforms: ***airborne*** (aircraft / drone), ***terrestrial*** (tripod / static) and ***mobile*** (vehicle-mounted) LiDAR.
- Output: dense point cloud; ***DSM*** (digital surface model) and ***DTM / bare-earth model*** by classifying ground returns; contours, 3-D city models.
- Advantages: very high accuracy (a few cm); works through gaps in vegetation (multiple returns); day or night; fast for large areas; detailed terrain in forests and in corridors.
- Limitations: costly equipment; large data volume to process; poor in heavy rain / fog; dark or reflective surfaces.
- Applications: flood modeling, highway and railway design, power line mapping, forest inventory, coastal and urban planning, archaeology, mining volume.

DIFFERENCE:

Drone survey = photographs (passive). LiDAR = laser pulses (active). Both give DEM / point clouds; LiDAR penetrates vegetation, photogrammetry cannot.

12. Worked Example – Deflection Angle Table

Using the earlier data (R = 300 m, D = 36 deg, chainage of T1 = 1092.52 m, T2 = 1281.02 m, peg interval 30 m). The first sub-chord = 1110 − 1092.52 = 17.48 m; last sub-chord = 1281.02 − 1260 = 21.02 m.

Peg	Chainage	Chord (m)	Deflection (min)	Cumulative
T1	1092.52	–	0	0 00
1	1110.00	17.48	100.15	1 40 09
2	1140.00	30.00	171.89	4 32 02
3	1170.00	30.00	171.89	7 23 55
4	1200.00	30.00	171.89	10 15 48
5	1230.00	30.00	171.89	13 07 41
6	1260.00	30.00	171.89	15 59 34
T2	1281.02	21.02	120.44	18 00 00 (= D/2)

CHECK:

Check: the final cumulative deflection must equal D/2 = 18 deg. If not, there is an arithmetic error – recheck the sub-chords.

13. Superelevation and Transition (short note)

- **Superelevation (cant):** raising the outer edge of the road / rail on a curve to balance the centrifugal force. e = V^2 / (127 R) for roads (V in km/h, R in m); cant for railways = G V^2 / (127 R).
- **Transition curve:** needed to introduce the radius gradually, to provide the superelevation gradually, and to avoid sudden lateral jerk. Length L = V^3 / (C R) (Shortt's formula).
- **Vertical curves:** parabolic curve joining two grades; ***summit*** (crest) curve when the gradient decreases, and ***sag*** curve when it increases. Rate of change of grade is kept small for comfort and sight distance.
- Typical grade change: length of a vertical curve L = N x S, where N = algebraic difference of the gradients, S = the sight distance factor.

14. Another Tacheometry Problem

A tacheometer set at P (R.L. of axis = 450.00 m), K = 100, C = 0. Staff held vertically at Q: vertical angle = -3 deg; readings: 1.200, 2.050, 2.900. Find D and R.L. of Q.

s = 2.900 - 1.200 = 1.700 m

D = 100 x 1.7 x cos^2 3 deg = 169.53 m

V = 100 x 1.7/2 x sin 6 deg = 8.88 m (depression)

R.L. of Q = 450.00 - 8.88 - 2.050 = 439.07 m

- Remember: for depression angle, subtract V and subtract the axial reading h from the R.L. of the axis.
- When the staff is held normal to the line of sight (instead of vertical): D = K s cos(theta) + C cos(theta) + h sin(theta); V = K s sin(theta) + C sin(theta).

15. Important Questions to Prepare

- Define a simple circular curve; derive the formulae for tangent length, length of curve and long chord.
- Explain the Rankine's deflection angle method of setting out a circular curve with the help of a table.
- Differentiate between compound and reverse curves; give their uses.
- Derive the distance and elevation formulae for the stadia method of tacheometry (inclined sight, staff vertical).
- Explain the principle and types of EDM; the advantages and applications of a total station.
- Write short notes on GPS, drone survey and LiDAR survey.

16. Quick Revision

- Simple curve: T = R tan(D/2); l = pi R D/180; L = 2R sin(D/2); chainage T1 = V - T.
- Deflection angle for a chord = 1718.9 c / R minutes; the total = D/2 (check).
- Tacheometry: D = K s cos^2 theta, V = (K s/2) sin 2 theta.
- EDM: D = ct/2; total station = theodolite + EDM + microprocessor; GPS needs minimum 4 satellites.
- Drone = camera images; LiDAR = laser point cloud.

SURVEYING

Handwritten Study Notes

Subject :	SURVEYING
Code :	23A01301T
Unit :	V - Photogrammetry Surveying

I. Introduction to Photogrammetry

Photogrammetry is the science and art of obtaining ***reliable measurements and maps from photographs*** . The word comes from photo (light), gramma (record) and metron (measure).

- Photographs are taken from the air (aerial) or from the ground (terrestrial); measurements on the photographs give the positions, shapes, sizes and heights of objects.
- Used for topographic mapping, contour maps, DEMs, road / railway / canal alignment, land use, forestry, geology, urban planning, military intelligence and archaeology.

Advantages

- Covers large areas quickly; the permanent record of the ground at an instant of time.
- Access to inaccessible and dangerous terrain; less fieldwork; economical for large projects.
- Data can be processed repeatedly; both planimetric and height data from the same photographs (stereo pairs).

Limitations

- Weather and light dependent; heavy initial equipment cost; needs ground control points; poor under dense forest and cloud.

2. Basic Concepts and Classification

Basis	Classification
Position of the camera	**Aerial** (from aircraft, drone, satellite) and **Terrestrial** (from ground stations)
Camera axis (aerial)	**Vertical** (tilt not more than 3 deg), **Tilted / near-vertical** (tilt 3 - 30 deg), **Oblique** (axis intentionally tilted: low oblique does not show the horizon; high oblique shows the horizon)
Equipment used	Analogue (film), analytical, **digital photogrammetry**
Purpose	Metric (measurement) and interpretative photogrammetry
Type of film	Panchromatic, infrared, colour

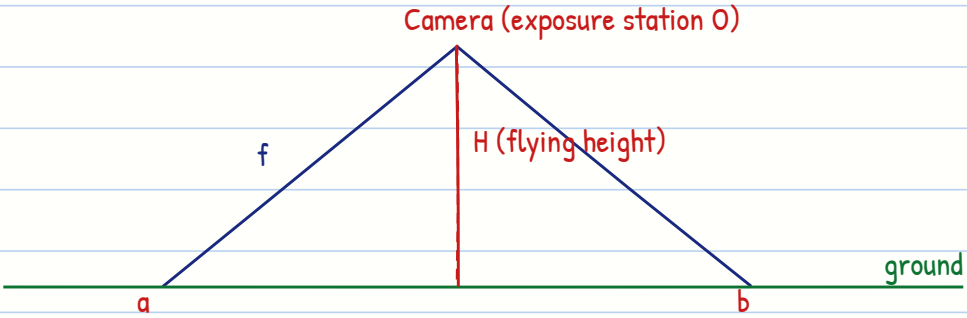


Fig 1: vertical aerial photograph geometry

Terms used on an aerial photograph

Term	Meaning
Principal point (p)	Foot of the perpendicular from the lens centre (rear nodal point) on the photo plane - the centre of the photograph
Nadir point (n)	Point where the plumb line through the exposure station meets the photograph
Isocentre (i)	Point on the photograph that lies on the bisector of the tilt angle
Exposure station (O)	Position of the camera lens at the time of exposure
Fiducial marks	Marks at the sides / corners of a photograph; their intersection = the principal point; used for measurements

Focal length (f)	Distance from the lens to the photographic plane (typically 152 mm, 210 mm, 88 mm)
Flying height (H)	Height of the exposure station above the datum (mean ground level)
Tilt (t)	Angle between the camera axis and the vertical
Parallax	Apparent displacement of an object due to change in the observer's position

3. Perspective Geometry of an Aerial Photograph

An aerial photograph is a ***central (perspective) projection*** : all the rays pass through the lens. A vertical photograph of a flat terrain is a scale reproduction of the ground.

Scale of a vertical photograph

Scale S = f / (H - h) = photo distance / ground distance
For flat ground (h = 0): S = f / H
Representative fraction RF = 1 : (H - h) / f

- h = elevation of the ground point above the datum; so the scale varies with the height of the terrain (higher ground -> larger scale, because it is nearer to the camera).
- ***Average scale*** uses the average terrain elevation h_avg: S_avg = f / (H - h_avg).
- If a map or a known ground distance is available: S = (photo distance ab) / (ground distance AB).

Ground length = photo length / S ; Area on ground = photo area / S^2

Example

A vertical photo is taken with f = 152 mm from H = 3200 m above datum. A point at h = 400 m elevation. Scale = 0.152 / (3200 - 400) = 1 / 18,421. An 8 cm line on the photograph = 8 x 18,421 / 100 = 1473.7 m on the ground.

Coordinates of the points on a vertical photo

- Photo coordinates (x, y) measured with the principal point as origin; the ground coordinates are:

$$X = x (H - h) / f \; ; \quad Y = y (H - h) / f$$

Ground length AB = sqrt[(XB - XA)^2 + (YB - YA)^2]

4. Relief Displacement

On a vertical photograph, the images of points above or below the datum are ***displaced radially*** from their true plan positions. This is ***relief displacement*** . Tall objects (towers, buildings, trees) appear to lean outwards from the principal point (the nadir).

$$d = r h / H = r' h / (H - h)$$

d = relief displacement; r = radial distance of the top of the object from the principal point;

h = height of the object; H = flying height above the base of the object

- Displacement is ***outward*** for a point above the datum and ***inward*** for a point below the datum.
- d increases with r (distance from the centre) and with h; d is ***zero at the principal point*** (r = 0) and zero for h = 0.
- Height of an object from the relief displacement: ***h = d H / r*** (the base and the top radial distances from the principal point are measured on the photograph).

USE:

Relief displacement can be used to find heights of tall objects (chimneys, towers); and it is also the basis of stereoscopic viewing and the height measurement. It decreases with higher flying height.

Example

A chimney top is 90 mm from the principal point on the photograph, and the relief displacement = 3.6 mm. Flying height above the base = 1800 m. Find the height of the chimney.

$$h = d H / r = 3.6 \times 1800 / 90 = 72 \text{ m}$$

5. Tilt Displacement

A ***tilted photograph*** has its optical axis inclined to the vertical by a tilt angle t. The photograph is no longer a scale model of the ground: the scale varies from side to side and the points are displaced ***radially from the isocentre*** .

- Tilt displacement is measured radially from the isocentre; it grows with the square of the radial distance and with the angle of tilt (approximately d_t = r^2 sin t cos phi / f for small tilts).
- Directions: points on the upward side of the isometric parallel are displaced ***away from*** the isocentre, on the lower side ***towards*** the isocentre; points on the isometric parallel (the line through the isocentre perpendicular to the principal line) are not displaced.

Tilt displacement (approx.): d_t = (r^2 / f) sin(t) cos(phi)
r = radial distance from the isocentre; phi = angle measured from the principal line

- Principal plane: vertical plane through the camera axis; principal line: its trace on the photograph; isometric parallel, axis of tilt, true horizon line - the three key lines of a tilted photograph.
- Tilt is removed by ***rectification*** (producing an equivalent vertical photo).

6. Terrestrial Photogrammetry

The photographs are taken from ***fixed stations on the ground*** with a ***phototheodolite*** (a camera mounted on a theodolite) or a metric camera. Used for close-range work: buildings, bridges, dams, mines, landslides, archaeology and accident sites.

- ***Phototheodolite:*** a camera + theodolite - gives the direction of the camera axis and the horizontal angle.
- ***Stereometric camera:*** two cameras fixed on a base bar of a known length (e.g. 1.2 m) - to give a stereo pair of the same area.
- Camera axes: (1) ***Horizontal / level*** and normal to the base line (***normal case***); (2) ***Parallel*** , (3) ***Convergent*** , (4) ***Divergent*** - the axes may be parallel or oriented at an angle.

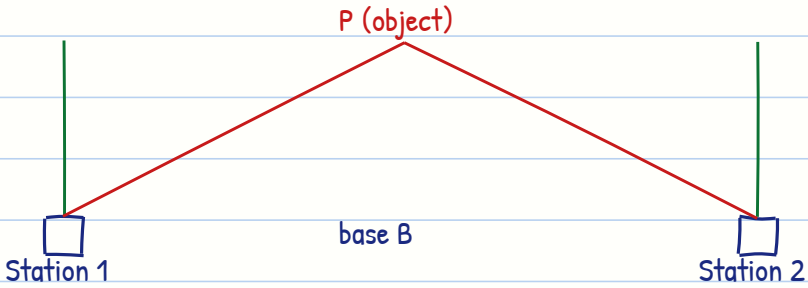


Fig 2: terrestrial stereo photographs (convergent pair)

Distance (Y) = (B x f) / (x_left - x_right) [normal case]

B = base length; f = focal length; the difference of x-coordinates = parallax p

- Distance from the camera: $Y = B f / p$; Height: $Z = Y y / f$; Lateral coordinate: $X = Y x / f$.
- Advantages: low cost, quick, exact records of the object for measurement later, and no contact with the object. Limitations: obstruction in view, shadow, hidden areas.

7. Flight Planning for Aerial Photography

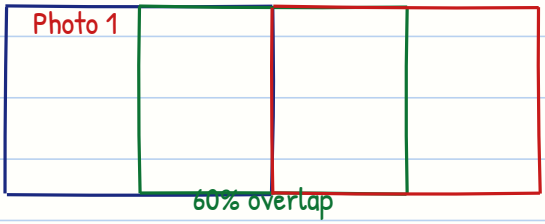
A good flight plan is essential for efficient and economical aerial mapping. The ***flight map*** is prepared on an existing map showing the flight lines, the exposure stations and the scale.

Factors considered

- Purpose and the scale of the final map; required accuracy; area, shape, terrain relief; camera focal length and the format (e.g. 23 x 23 cm); weather and season (sun angle 30-40 deg, no clouds, snow, leaves).
- Flight height, speed of the aircraft, overlap, number of strips and the number of photographs.

Overlaps

- ***Forward (longitudinal) overlap:*** 60 % (55-65 %) between the successive photographs in a strip - required for stereo vision.
- ***Side overlap (lateral):*** 20 - 30 % (typically 25 - 30 %) between the adjacent strips - avoids gaps.



forward overlap in a strip

Fig 3: forward overlap of photographs

Design formulae

Ground coverage of one photo: L = l x S_num (l = photo side; S_num = 1/scale number)

Flying height H = f x S_num (above mean ground level)

Effective ground coverage (forward) = L (1 - Pf/100); (side) = L (1 - Ps/100)

Air base B = L (1 - Pf/100) ; Distance between flight lines W = L (1 - Ps/100)

Number of photos per strip = (strip length / B) + 1 (+ 2 extra at the ends)

Number of strips = (width of the area / W) + 1

Time interval between exposures = B / (speed of the aircraft)

Example

Area 12 km x 8 km; scale 1:15,000; photo 23 cm x 23 cm; f = 152 mm; Pf = 60%, Ps = 30%; aircraft speed 200 km/h.

L = 0.23 x 15,000 = 3450 m

H = 0.152 x 15,000 = 2280 m

B = 3450 x 0.4 = 1380 m ; W = 3450 x 0.7 = 2415 m

Photos per strip (12 km) = 12000/1380 + 1 = 8.7 + 1 = 10 (+2) = 12

Strips = 8000/2415 + 1 = 3.3 + 1 = 4.3 ~ 5

Time interval = 1380 / (200000/3600) = 24.8 sec

- Total photos = 12 x 5 = 60. Always allow extra photos for the ends of each strip, turning and wastage.

8. Stereoscopy

Stereoscopy is the science of seeing in three dimensions using two overlapping photographs of the same area taken from different positions (a **stereo pair**). Each eye sees one photograph, and the brain gives a 3-D image of the terrain (stereo model).

- **Stereoscopic vision:** the eyes see objects from slightly different directions; the **parallactic angle** gives the depth perception. Human eye base is 6.5 cm (about 65 mm).

Stereoscopes

Type	Description
Lens (pocket) stereoscope	Two simple lenses on a frame, about 5 cm apart, focal length 10 cm; small field; cheap and portable
Mirror stereoscope	Two large mirrors and two small ones; full overlap of 23 cm photos can be seen; with binoculars for magnification
Scanning / zoom stereoscopes	Higher magnification (zoom), used for the detailed interpretation
Anaglyph / polarised system	Red and blue (or polarised) images are separated by the coloured (polarised) glasses

Parallax and the height measurement

Stereoscopic (x-) parallax p of a point = the algebraic difference of the distances of its two images from the respective principal points, measured along the flight line (base line).

Absolute parallax of a point: $p = x_{left} - x_{right}$ (x in the photo coordinates measured on the flight line)

Elevation of a point: $h = H - (B f / p)$

Difference in elevation: $dh = (H - h_A)\, dp / (p_a + dp) = (H - h_A)\, dp / p_b$

- **Parallax bar (stereometer)** is used to measure the differences in parallax accurately (to 0.01 mm) while looking in the stereoscope: dp is the difference in the bar reading between two points.

- **Air base B** = ground distance between the two exposure stations; the photo base b = the average distance between the principal point and the conjugate principal point on the photographs: p = b for the point at the principal point.

> **NOTE:**
> Higher points have larger parallax, lower points have smaller parallax. Heights of objects from the parallax difference: h = (H x dp) / (p_base + dp) where p_base is the parallax at the base of the object.

Example

H = 2000 m above ground. Photo base b = 90 mm. The parallax difference between the top and the base of a tower = 1.2 mm. Height of tower:

$$h = H\,dp / (p_{base} + dp) = 2000 \times 1.2 / (90 + 1.2) = 26.3 \text{ m}$$

9. Ground Control Extension for Photographic Mapping

To plot photographs to a map scale we need ***ground control points (GCPs)*** - points identifiable on the photograph whose positions (X, Y, Z) are known from the ground survey (GPS, total station, levelling). Surveying every control point on the ground is costly. So control is ***extended*** from a few ground points by ***phototriangulation*** (aerial triangulation).

- ***Horizontal control:*** (X, Y) at least 2-3 points per model; ***vertical control:*** (Z) at least 3 points.
- ***Aerial triangulation*** gives the coordinates of the additional control points (***pass points / tie points***) in the strip or block from the measurements on the photos. It reduces the field work by a very large amount.

Types of aerial triangulation

Method	Principle
Radial triangulation (graphical / slotted templet)	Uses the property that angles at the principal point are true on a vertical photo (radial lines); intersection of rays gives horizontal positions
Analogue (instrumental) triangulation	Using stereo-plotting instruments, strips of the models are oriented successively

Analytical triangulation	Mathematical (least squares) adjustment of the strip / block with the photo coordinates; the ***bundle adjustment*** is the most accurate
GPS / INS supported triangulation	Exposure station coordinates and orientation from the onboard GPS and IMU reduce the ground control needs

Radial triangulation

Based on the principle that ***on a vertical photograph, the angles measured at the principal point (radial angles) are true horizontal angles*** (displacements due to tilt and relief are radial from the centre and do not change the angle).

- ***Graphical (paper templet) method:*** the principal points and the pass points are marked on each photograph; a transparent overlay (templet) is made with the radial lines from the principal point (centre) to each marked point. The templets of adjacent photographs are joined using the common points and the ground control, and the intersections give the unknown positions.
- ***Slotted templet method:*** the rays are represented by slots cut in a cardboard (or celluloid) templet; the studs fix the points. Assembled like a Meccano set to the ground control. Fast; used for large blocks.
- ***Mechanical (Arundel) method:*** like the slotted templet; ***Radial line plotter*** .
- Steps: (1) Select and mark the principal points, conjugate principal points, pass points and control points. (2) Draw radial lines. (3) Intersect the rays from the adjacent photos to get new points. (4) Scale and orient the strips to the ground control and adjust.

> ***NOTE:***
> Radial triangulation gives ***only plan (X, Y) positions*** ; elevations are got by the spirit levelling, or by the parallax method or the stereo-plotter.

10. Photographic Mapping

Maps can be produced from the aerial photographs by several methods, depending on the accuracy, cost, and the equipment available.

(a) Mapping using paper prints

- Photographic prints (contact prints at the photo scale) are used directly after ***rectification*** and the ***ratioing*** to the map scale; details are ***traced*** on the transparent overlay and the map is compiled with control.
- Planimetric map by the ***radial line method / sketchmaster / vertical sketchmaster / reflecting projector*** that enable the transfer of detail from the photo to the base map.
- Limitation: the displacements due to relief and tilt remain, so the accuracy is limited and the method suits flat terrain and small-scale maps.

(b) Mapping using stereo-plotting instruments

Stereo-plotters (stereo-plotting instruments) reconstruct the 3-D model from the overlapping photographs and let the operator trace the planimetric details and contours directly in the model with a floating mark. They give the ***accurate planimetric and contour maps*** .

Class	Examples / working
Direct optical projection	Multiplex, Balplex: diapositives projected optically; the floating mark on a platen is moved over the model
Mechanical projection	Wild A-series, Kelsh plotter: spatial rods represent the projected rays; model scale adjusted by the base
Analytical plotters	Computer-controlled; photo coordinates measured and mathematically transformed to the ground coordinates (high precision)
Digital photogrammetric workstations (softcopy)	Digital images on a computer; automated matching to generate the DEM, orthophotos, contours, 3-D models

- Stages: ***interior orientation*** (restoring the camera geometry), ***relative orientation*** (removing the y-parallax to form the model) and ***absolute orientation*** (scaling and levelling the model using the ground control).
- Floating mark: the operator moves a mark in x, y, z so that it appears to rest on the model; the instrument records / plots the position -> planimetry and contours.

(c) Mosaics

A **mosaic** is an assembly of overlapping aerial photographs (trimmed and joined) to form a continuous picture of the area.

- **Controlled mosaic:** made from rectified and ratioed photos laid to the ground control; accurate; scale is uniform.
- **Uncontrolled mosaic:** made by matching the details of the adjacent photos, without any control; quick and cheap; not accurate.
- **Semi-controlled mosaic:** a combination – the control is used for some portions.
- Methods: the central portion of each photo is used (to reduce the displacements); cut along the patterns / feathered edges; photographed to the final scale.
- Uses: quick view / planning, interpretation of the whole area, as a map substitute, base for the thematic mapping.

(d) Map substitutes

Products made from the aerial photographs that can be used **instead of a line map** , when quick and economical information is needed.

- **Photomap:** a mosaic or photograph with the grid, names and the symbols added – map substitute for rapid planning.
- **Orthophotograph / orthophoto map:** photograph in which the relief and tilt displacements have been removed (differential rectification using a DEM) – so it has a uniform scale like a map; with contours it is an orthophoto map.
- **Mosaic (controlled / uncontrolled) maps** and **stereo-pairs** (for the interpretation).

> **KEY:**
> Orthophoto = photograph + map. It has the details of a photo and the accuracy of a map (uniform scale). Digital orthophotos are produced by the softcopy workstations / drone processing.

II. Photo Interpretation (brief)

- Identification of objects from their **tone, texture, shape, size, shadow, pattern, location and association** .
- Uses in geology, forestry, agriculture, hydrology, urban and transport planning, and disaster assessment.

12. Modern Trends

- Digital cameras, ***drones*** (UAV photogrammetry), ***GNSS-INS*** (direct georeferencing), ***LiDAR*** , satellite stereo (Cartosat, SPOT, WorldView) for DEM and mapping; automatic image matching and ***Structure-from-Motion*** .
- Software products: Pix4D, Agisoft Metashape, ERDAS, Trimble Inpho – produce orthomosaics, DSM and 3-D models in a short time.

13. Important Questions to Prepare

- Define photogrammetry. Explain the classification of photographs and the terms on a vertical photograph.
- Derive the expression for the scale of a vertical photograph and the relief displacement.
- Explain the tilt displacement and the effect of the tilt on a photograph.
- What are the factors in the flight planning? Calculate the number of photos for a given area.
- Describe the stereoscope types, the parallax bar and the use for finding the heights.
- Explain the radial triangulation methods. Write notes on mosaics and map substitutes.

14. Quick Revision

- Scale = f/(H – h). Relief displacement d = r h / H. Overlaps: 60 % forward, 30 % side.
- Height from the parallax: dh = (H – h) dp / (p + dp). Air base B = L (1 – Pf/100).
- Radial triangulation uses angles at the principal point which are true on a vertical photo.
- Stereo-plotter orientations: interior, relative, absolute. Orthophoto removes the relief and tilt displacements.